Coupling of the JOREK and DREAM codes for accurate and efficient runaway electron simulations in large tokamak devices

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Coupling of the JOREK and DREAM codes for accurate and efficient runaway electron simulations in large tokamak devices





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Populärvetenskaplig sammanfattning

Fusionsenergi, samma process som driver solen, skulle kunna bli en nästan outtömlig och ren energikälla. För att uppnå fusionsreaktionen på jorden måste temperaturen av partiklarna vara högre än i solens kärna, och därför används ett så kallat plasma, som måste inneslutas med hjälp av starka magnetfält. Detta görs i en torusformad reaktor, så kallad tokamak. Eftersom plasmat är ostabilt kan ibland kontrollen förloras på några tusendels sekunder. En sådan händelse kallas disruption.

Under en disruption kan en del av elektronerna i plasmat accelereras till hastigheter nära ljusets. Dessa så kallade skenande elektroner kan bilda en intensiv stråle som, om den träffar reaktorns väggar, kan orsaka allvarliga skador. För att kunna bygga säkra fusionsreaktorer är det därför viktigt att förstå hur skenande elektroner uppstår och hur de beter sig.

Eftersom experiment i stora tokamaker är både dyra och svåra att genomföra spelar datorsimuleringar en central roll. Det finns redan avancerade simuleringskoder för detta, men de har olika styrkor och svagheter. Koden JOREK ger en detaljerad tredimensionell bild av hur plasmat rör sig, men att följa elektronerna i detalj under en hel disruption blir alldeles för beräkningstungt. Koden DREAM kan däremot effektivt beräkna hur de skenande elektronerna utvecklas över långa tider, men saknar den fullständiga bilden av plasmats rörelser.

I detta projekt har de två koderna kopplats ihop så att de kan utnyttja varandras styrkor. Under simuleringens gång beskriver JOREK hur plasmat som helhet utvecklas, medan DREAM beräknar hur de skenande elektronerna uppstår och accelereras. De två programmen skickar information till varandra under hela förloppet, vilket gör det möjligt att simulera disruptioner över längre tider än tidigare utan att förlora viktig information om plasmat.

Den nya modellen har jämförts med en simulering av en disruption i EU-DEMO, en planerad europeisk demonstrationsreaktor, och visar god överensstämmelse under disruptionens tidiga skede. Arbetet utgör ett första steg mot mer realistiska förutsägelser av skenande elektroner, och därmed mot säkrare fusionsreaktorer i framtiden.

Abstract

On the path to creating magnetic confinement fusion devices, such as the tokamak, to demonstrate net energy gain, several obstacles remain in the way. One of the major issues for tokamaks are so called disruptions which consist of the plasma losing its confinement over a millisecond time scale. During these disruptions a fraction of the plasma electrons can get accelerated to relativistic energies, referred to as runaway electrons (REs), and are a significant threat to the plasma facing components. The study of RE formation and dynamics is therefore crucial for the safety and integrity of future fusion devices. To study the formation and dynamics of REs, several simulation codes exist, ranging from fast lower-dimensional codes to high fidelity ones. A leading code to study RE formation and dynamics during disruptions is the DREAM (Disruption Runaway Electron Analysis Model) which is a one-dimensional fluid-kinetic flux-surface averaged code. For more realistic simulations, the JOREK 3D nonlinear extended magnetohydrodynamic and hybrid fluid-kinetic code is widely used. Both codes offer the possibility of simulating the REs kinetically, i.e., keeping track of the phase-space distribution of the particles, but using two different models: DREAM by evolving the distribution function of the particles and JOREK by evolving markers representing the distribution function in a Monte Carlo approach. The highest-fidelity kinetic model is the 3D particle-in-cell kinetic model offered by JOREK, but its heavy computational cost makes it unsuitable for the very long timescale simulations needed to cover a full disruption in a large tokamak device. On the other hand, DREAM lacks the full three-dimensional picture which JOREK offers. For this reason, a new model allowing kinetic simulations over long time scales while retaining accurate information about the background plasma is needed.

In this project, a new simulation capability for RE generation and dynamics in tokamak plasmas has been developed. By coupling the JOREK and DREAM codes, a new way of performing high-fidelity kinetic simulations over long disruption time scales has been established. The idea of this coupling is to trace the background plasma dynamics in JOREK while DREAM evolves the electron population kinetically and letting both codes communicate this information to each other during the simulation process. The model is validated against a JOREK disruption simulation of the EU-DEMO tokamak, where a good qualitative and quantitative agreement for the early current-quench dynamics.

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List of symbols

Symbol	Definition
ψ	Poloidal magnetic flux
$\hat{\psi}$	Normalized poloidal flux
ψ_{ax}	Poloidal magnetic flux at magnetic axis
ψ_{bnd}	Poloidal magnetic flux at last closed flux surface
$(R_{\mathrm{ax}}, Z_{\mathrm{ax}})$	Position of the magnetic axis in the (R, Z) plane.
θ	Poloidal angle
φ	Toroidal angle
$\boldsymbol{E}$	Electric field
$\boldsymbol{B}$	Magnetic field
$\boldsymbol{v}$	Particle velocity
c	Speed of light in vacuum
q	Particle charge
m	Particle mass
$\boldsymbol{n}$	Particle density
$\boldsymbol{V}$	Mean plasma velocity
$\boldsymbol{T}$	Plasma temperature
p	Magnitude of the particle momentum or plasma pressure
ξ	Pitch of the particle
E_c	Connor-Hastie electric field
I_p, I_{tot}	Total plasma current
I_{re}	Runaway electron current
ϵ_0	Vacuum permittivity
μ_0	Vacuum permeability
e	Electron charge
$\ln \Lambda$	Coulomb logarithm
ρ	Plasma density
η	Plasma resistivity
$\boldsymbol{J}$, $\boldsymbol{J_r}$	Total plasma and runaway current densities in JOREK
$\boldsymbol{j}$, $\boldsymbol{j_r}$	Toroidal total plasma and runaway current densities in JOREK
v	Poloidal flow potential
ω	Toroidal vorticity
T_e	Electron temperature
n_e	Free electron density
$E_{\parallel}$	Parallel electric field
Z_{eff}	Effective charge of the plasma
B_{min}	Minimum magnetic field strength on a flux surface
f_{hot}	Distribution function of the hot electron population

Symbol	Definition
f_{re}	Distribution function of the runaway electron population
j_{Ω}	Ohmic current density in DREAM
j_{hot}	Hot electron current density in DREAM
j_{re}	Runaway electron current density in DREAM
N_r	Number of radial grid cells in DREAM
n_{ψ}	Number of flux surfaces the magnetic equilibrium is evaluated on in JOREK
n_{θ}	Number of poloidal angle values in the DREAM numerical radial grid
t_{norm}	JOREK normalization time

List of abbreviations

Abbreviation	Meaning
CQ	Current quench
DREAM	Disruption Runaway Electron Analysis Model
EU-DEMO	European demonstration fusion power plant
FEM	Finite Element Method
GC	Guiding Center
MCF	Magnetic Confinement Fusion
MHD	Magnetohydrodynamics
MUSCLE3	Multiscale Coupling Library and Environment, version 3
PFC	Plasma Facing Component
PIC	Particle-in-cell
RE	Runaway electron
TQ	Thermal quench
VDE	Vertical Displacement Event
yMMSL	YAML-based Multiscale Modelling and Simulation Language

Chapter 1

Introduction

1.1 Magnetic confinement fusion in tokamaks and the challenge of disruptions

Nuclear fusion energy is considered as a promising approach to solving the world energy challenge by providing a nearly limitless and clean source of energy [1]. Fusion combines two atomic nuclei into one heavier nucleus, releasing a significant amount of energy in the process, and it is the mechanism which powers all the stars of the universe. To achieve fusion on Earth, Magnetic Confinement Fusion (MCF) is a particularly promising approach. For the fusion reaction to happen, the fusion fuel has to be hot enough and thus be in the form of a plasma. MCF confines plasma at temperatures exceeding 100 million kelvin using magnetic fields. Among MCF concepts, the tokamak has demonstrated the highest energy and power outputs, making it the leading design under investigation [2]. Tokamaks utilize both the toroidal and poloidal magnetic field to confine the plasma in a torus shaped domain. This is necessary to keep the plasma away from the walls to avoid melting of these. To do this, an induced electric field needs to drive a high toroidal current through the plasma.

This toroidal current in a tokamak, however, gives rise to several issues that are predicted to become more severe in high-power devices. Upon loss of confinement of the plasma due to instabilities or system malfunctions, the machine may undergo a so-called disruption. The detailed description of the disruption process is given in Chapter 2. They are a type of operational failure, which can lead to very large heat loads and mechanical stress on the structures of the tokamak. Additionally, during disruptions, highly energetic electrons known as Runaway Electrons (REs) can be formed [3]. These particles can deposit their energy in a localized manner on the wall structures of the tokamak, known as Plasma Facing Components (PFCs), and can lead to severe melting or even explosion of the PFCs [4].

Several experimental tokamaks are in current use around the world, such as WEST [5] in France, ASDEX Upgrade [6] in Germany or TCV [7] in Switzerland. Future high performance tokamaks are under construction such as ITER [8] which is being built in France, or being planned such as SPARC [9], BEST [10], CFETR [11], STEP [12], or a European first of a kind reactor [13]. Several of these tokamaks aim to demonstrate net energy production. The issue with these high-performance tokamaks is that the risk of damage from disruptions, notably runaways, scales unfavorably with the plasma current. Indeed, the multiplication of REs by large-angle collisions during the Current Quench (CQ), the avalanche mechanism, scales exponentially with the plasma current [3]. In fact, ITER is built for a maximum current of 15 MA whereas, for example, ASDEX Upgrade has

a maximum current of 2 MA. Consequently, avalanching is not very efficient in presently operating tokamaks, while it will be extremely strong in future high current tokamaks. For this reason it is difficult to draw conclusions from present experiments and extrapolate to future machines. As a consequence, analytical and numerical modeling are highly useful tools to validate against present day experimental results and to make predictions for future machines.

Due to the nonlinearity, multi-physics nature, and geometrical complexity of plasma dynamics inside a fusion reactor, analytical descriptions are challenging and thus numerical simulations have become key tools to predict the plasma behaviour in tokamaks. To study runaway electron dynamics in a tokamak plasma, both very short and very long time- and length scales need to be simulated. For this purpose, a range of simulation codes exist: from high-fidelity, three-dimensional codes such as JOREK [14, 15], which can resolve the coupled dynamics of REs and the background plasma but at prohibitive computational cost over a full disruption, to computationally efficient, lower-dimensional codes such as DREAM [16], which evolve the RE phase-space distribution over long disruption timescales [17, 18] but without a self-consistent three-dimensional plasma description. This thesis introduces, for the first time, a coupled JOREK-DREAM model that bridges this gap, developed entirely within the scope of this work. The full motivation for this coupling, and how it fits into the broader hierarchy of RE simulation tools, is discussed in Section 3.1, once the necessary theoretical background has been introduced.

1.2 Aim of the thesis and outline

Concretely, this thesis aims to establish the coupling of the JOREK and DREAM codes to allow for kinetic simulations of RE dynamics over long disruption time scales where the 3D plasma effects do not play a crucial role and thus the high-fidelity treatment of the plasma and RE dynamics as offered in the computationally much more expensive JOREK standalone simulations is not needed. The aim is to do so by letting JOREK evolve the plasma state, while DREAM evolves the REs kinetically, and exchange the relevant quantities between both codes during the course of the simulation so that jointly, JOREK and DREAM can evolve the plasma and the REs accurately and self-consistently.

The thesis is divided as follows. Chapter 2 introduces the background, the theory and the simulation tools needed for the work in this thesis. Thereafter, Chapter 3 presents the main work of this thesis, which is the methodology used in the coupling of JOREK and DREAM. Chapter 4 then presents numerical verification and benchmarking of the coupled model as well as some initial results obtained using it. Finally, Chapter 5 finishes this thesis with a summary of the key new capabilities that can be obtained using the JOREK-DREAM coupled model and describes future planned improvements to the model and novel research directions enabled by these capabilities.

1.3 Use of AI

AI has been used as a tool to find references and clarify some concepts found in the litterature.

It has also been used to help generate the code for the following figures in this manuscript: Figure 2.1, Figure 2.2, Figure 2.3, Figure 2.4, Figure 2.6, Figure 2.8, Figure 3.1, Figure 3.2, Figure 3.3 and Figure 3.4.

For coding, AI has been used to help debug errors in the code and to help generate plotting scripts.

Chapter 2

Background and theory

This chapter presents the theoretical and conceptual background required to understand the physics and the numerical methodology developed throughout this thesis. Section 2.1 starts by introducing the physical concepts on which this thesis is based. Sections 2.2 and 2.3 present the physical and numerical models used in the JOREK and DREAM codes which are aimed to be coupled in this thesis.

2.1 Tokamaks and plasma physics

To understand the work done in this thesis, a few plasma physics concepts are necessary, these will now be detailed. Subsection 2.1.1 presents the tokamak concept and its geometry. Then, Subsection 2.1.2 presents the basic properties and equations of a plasma needed to understand the kinetic plasma description used in DREAM and the magnetohydrodynamic (MHD) framework employed in JOREK. Finally, Subsection 2.1.3 describes the phenomenology of disruptions and of RE generation.

2.1.1 The tokamak and its geometry

For the fusion reaction to be possible, the fusion fuel must be in the form of a plasma at high enough temperatures. Plasma is often characterized as the fourth state of matter. It is defined as being a macroscopically neutral gas containing charged particles that exhibit collective behaviour due to the long range electromagnetic forces of the charged particles. A tokamak is a toroidal device designed to confine a high-temperature plasma using magnetic fields. Plasma confinement is achieved through the combination of two magnetic field components: a strong toroidal magnetic field, generated by the toroidal field coils, and a weaker poloidal magnetic field. The toroidal magnetic field guides charged particles around the torus, while the poloidal magnetic field twists the magnetic field lines into helical trajectories. The poloidal magnetic field is generated primarily by the toroidal plasma current, which is induced by a toroidal electric field produced by the central solenoid. In addition, the poloidal field coils surrounding the vacuum vessel provide the external magnetic fields required for plasma shaping, positioning, and equilibrium control. The combined toroidal and poloidal magnetic fields produce helical magnetic field lines that the charged particles closely follow, allowing the plasma to remain confined away from the vessel walls. Magnetic confinement is essential to prevent the plasma, whose temperature can exceed 10^8 K, from coming into direct contact with the plasma-facing components (PFCs), thereby avoiding excessive heat loads and material damage [19]. A simplified illustration of a tokamak is shown in Figure 2.1.

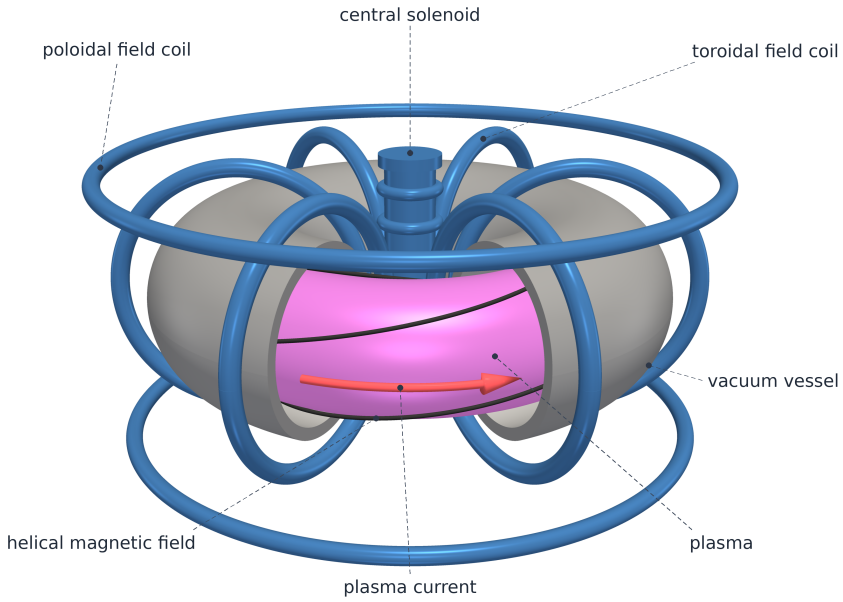


Figure 2.1: Simplified illustration of a tokamak reactor. The confinement of the plasma is ensured via a helically shaped magnetic field generated via external coils and by driving a current through the plasma.

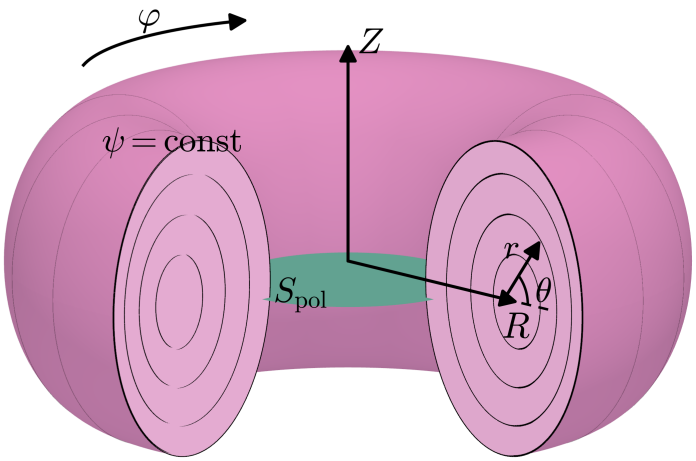


Figure 2.2: Illustration of the magnetic geometry of a tokamak. The green disk represents the surface over which the magnetic field is integrated to obtain the poloidal magnetic flux ψ. The magnetic field lines follow surfaces of constant poloidal flux. The geometry can be parametrized using cylindrical coordinates (R, Z, φ), poloidal coordinates (r, θ, φ), magnetic coordinates (ψ, θ, φ) or $(\hat{\psi}, \theta, \varphi)$.

The magnetic field lines lie on a family of nested closed surfaces known as magnetic flux surfaces [20], which are surfaces of constant poloidal magnetic flux $\psi = \int_{S_{\text{pol}}} \mathbf{B} \cdot d\mathbf{S}$, where S_{pol} is the green surface in Figure 2.2. The plasma geometry is conveniently described in the cylindrical coordinate system (R, Z, φ), where R is the major radial coordinate measured from the axis of symmetry of the tokamak, Z is the vertical coordinate, and φ is the toroidal angle. A radial coordinate r is then introduced as a label for the flux surfaces and is chosen to be constant on each surface. Another poloidal flux coordinate can be introduced and is the normalized poloidal flux $\hat{\psi} = \frac{\psi - \psi_{\text{ax}}}{\psi_{\text{bnd}} - \psi_{\text{ax}}}$, where ψ_{ax} is the flux at the magnetic axis and ψ_{bnd} is the flux at the last closed flux surface. This ensures that $\hat{\psi}$ is zero at the magnetic axis and one at the last closed flux surface. Furthermore, the innermost flux surface at $r = 0$ is called the magnetic axis and is characterized by its position $(R_{\text{ax}}, Z_{\text{az}})$ in the (R, Z) plane. The cross section of a plasma is typically not circular and is most often elongated and of almost triangular shape, as illustrated in Figure 2.2.

Owing to the nested structure of the magnetic flux surfaces, it is convenient to describe the plasma using poloidal coordinates (r, θ, φ) or magnetic coordinates (ψ, θ, φ), where ψ labels the flux surfaces, θ is the poloidal angle, and φ is the toroidal angle. When neglecting guiding-center drifts, charged particles follow the magnetic field lines and therefore remain on surfaces of constant ψ. The poloidal angle θ specifies the position around a flux surface in the poloidal plane and is conventionally measured from the outboard midplane with respect to the magnetic axis. The toroidal angle φ specifies the position around the torus measured from a chosen toroidal reference direction.

2.1.2 Basic plasma physics

Particle orbits in magnetic fields

In the presence of electric and magnetic fields, particles experience the Lorentz force given by:

$$\boldsymbol{F} = q(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}), \tag{2.1.1}$$

where $\boldsymbol{E}, \boldsymbol{B}, \boldsymbol{v}$ and q represent the electric and the magnetic fields, the particle velocity and the charge of the particle, respectively. The Lorentz force guides charged particles into helical orbits that wind around the magnetic field lines, which themselves wrap around the torus. This gyromotion occurs at the cyclotron frequency $\Omega = qB/\gamma m$ and radius $r_L = v_\perp/\Omega$, where $v_\perp$ is the particle velocity perpendicular to $\boldsymbol{B}$ and m is the mass of the particle. The relativistic factor γ is defined as $\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$ where v is the magnitude of the particle velocity $\boldsymbol{v}$ and c is the speed of light in vacuum. Since the toroidal field of a tokamak is stronger close to the torus centre, the field strength experienced by a particle varies as it moves along its orbit, with important consequences for its motion [21].

It is convenient to describe a particle's velocity by its magnitude and its direction relative to the local magnetic field. The particle momentum is defined as $\boldsymbol{p} = \gamma m \boldsymbol{v}$, with magnitude $p = |\boldsymbol{p}|$, and the pitch, or pitch-angle cosine, is defined as

$$\xi \equiv \frac{v_\parallel}{v} = \frac{p_\parallel}{p}, \tag{2.1.2}$$

where $v_\parallel = \boldsymbol{v} \cdot \boldsymbol{B}/B$ is the velocity component parallel to $\boldsymbol{B}$ and where B denotes the magnetic field strength. A pitch of $\xi = 1$ ($\xi = -1$) corresponds to a particle streaming exactly parallel (antiparallel) to $\boldsymbol{B}$.

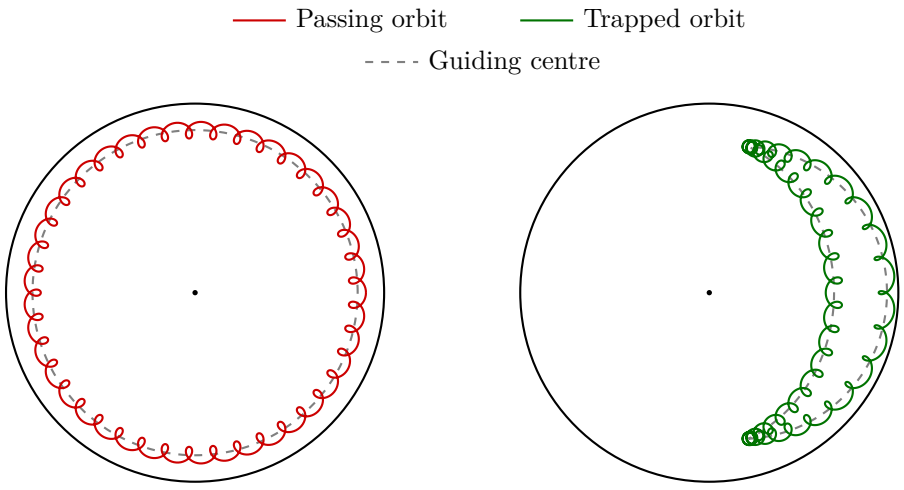
A key result is that the magnetic moment $\mu \equiv mv_\perp^2/2B$ behaves as an adiabatic invariant, remaining practically constant over the course of a particle's orbit [19]. Now letting $v_\parallel$ denote the particle velocity in the direction of $\boldsymbol{B}$, if the kinetic energy is also conserved, this implies (in the non-relativistic limit)

$$W_{\text{kin}} = \frac{mv^2}{2} = \frac{mv_\parallel^2}{2} + \frac{mv_\perp^2}{2} = \frac{mv_\parallel^2}{2} + \mu B = \text{const.} \tag{2.1.3}$$

As a particle moves into a region of stronger field, μ staying fixed forces $v_\parallel$ to decrease. If the field becomes strong enough, $v_\parallel$ is driven to zero and changes sign, causing the particle to bounce back the way it came. Whether this happens depends on the particle's initial pitch, i.e. the ratio $\xi_{\text{ini}} = v_{\parallel,\text{ini}}/v_{\text{ini}}$: particles with a large enough ratio are reflected and become trapped, bouncing between two poloidal angles and tracing out a characteristic banana-shaped orbit when projected onto a poloidal cross-section; particles with a smaller ratio complete a full poloidal transit unobstructed and are called passing particles. For a particle with initial pitch ξ_{ini} at a starting location where the magnetic field strength is B_{ini}, the condition for the particle being trapped is [20]:

$$\frac{B_{\text{ini}}}{B_{\text{max}}} < 1 - \xi_{\text{ini}}^2, \tag{2.1.4}$$

where B_{max} is the maximum magnetic field strength along the particle's orbit. An illustration of passing and trapped orbits is given in Figure 2.3.



(a) Passing orbit (b) Trapped orbit

Figure 2.3: Illustration of the two types of particle orbits in a tokamak plasma projected onto a poloidal cross section. (a) The passing orbit. (b) The trapped orbit.

The kinetic description

The most precise plasma model would be to model all separate particles including the forces acting on them, however this will quickly become computationally untractable due to the very large number of particles in a tokamak plasma (in the order of 10^{22} particles). For this reason, particles are modelled collectively. One of the most common model for this is called the kinetic model [21]. In this model the plasma is described using the distribution function of the particles $f(\boldsymbol{r}, \boldsymbol{v}, t)$. This distribution function is defined such

that $f(\boldsymbol{r}, \boldsymbol{v}, t)d\boldsymbol{r}d\boldsymbol{v}$ represents the expected number of particles in the infinitesimal six-dimensional position and velocity volume element $d\boldsymbol{r}d\boldsymbol{v}$.

The evolution of the distribution function is governed by the Boltzmann equation:

$$\frac{\partial f}{\partial t} + \boldsymbol{v} \cdot \nabla f + \frac{\boldsymbol{F}}{m} \cdot \nabla_v f = C[f], \tag{2.1.5}$$

where the force $\mathbf{F}$ is the Lorentz force (2.1.1) and $C[f]$ is a collision operator representing the collisions of the plasma particles with each other. There are several models for the collision operator, however in this thesis only the Fokker-Planck operator is used, which is given in advection-diffusion form by:

$$C[f] = \frac{\partial}{\partial \mathbf{v}} \cdot \Big[- \boldsymbol{A} f + \mathbb{D} \cdot \frac{\partial f}{\partial \boldsymbol{v}} \Big], \tag{2.1.6}$$

where $\boldsymbol{A}$ is an advection vector and $\mathbb{D}$ a diffusion tensor [22]. The DREAM code evolves the dynamics of the electrons in the plasma using the Boltzmann equation coupled to the Fokker-Planck collision operator, which is usually referred to as the Fokker-Planck equation. This will be detailed in Section 2.3.

It is possible to obtain macroscopic quantities by taking velocity moments of the distribution function. The three most important ones are:

$$\text{Particle density: } n(\boldsymbol{r}, t) = \int f(\boldsymbol{r}, \boldsymbol{v}, \boldsymbol{t}) d\boldsymbol{v} \tag{2.1.7}$$

$$\text{Mean velocity: } \boldsymbol{V}(\boldsymbol{r}, t) = \frac{1}{n(\boldsymbol{r}, t)} \int \boldsymbol{v} f(\boldsymbol{r}, \boldsymbol{v}, \boldsymbol{t}) d\boldsymbol{v} \tag{2.1.8}$$

$$\text{Temperature: } T(\boldsymbol{r}, t) = \frac{2}{3n(\boldsymbol{r}, t)} \int \frac{m}{2} |\boldsymbol{v} - \boldsymbol{u}(\boldsymbol{r}, t)|^2 f(\boldsymbol{r}, \boldsymbol{v}, \boldsymbol{t}) d\boldsymbol{v}. \tag{2.1.9}$$

These are respectively called the zeroth, first and second order moments of the distribution function.

From the kinetic to the fluid description

Another way of modelling plasmas is in a fluid (or a continuum) manner. The fluid model simplifies the kinetic description by taking velocity moments of the Boltzmann equation, thereby averaging out the velocity dependence and retaining only quantities that depend on position and time. It is thus possible to describe the plasma through the macroscopic quantities introduced above, which are related to each other through fluid equations obtained by taking velocity moments of the Boltzmann equation. The advantage of this approach is that it greatly reduces the complexity of the kinetic model, as the dimensionality of the problem is reduced, but lose details that are not captured by the taken velocity moments. The MHD model, which is used in the JOREK code and will be detailed in Section 2.2 is such a fluid model. It is also possible to combine fluid and kinetic approaches, where certain particle species are described in the fluid framework and others in the kinetic framework. This is the case for the high fidelity hybrid fluid-kinetic RE model inside JOREK [23], and it is also the case for the JOREK-DREAM model established in this thesis, where the background plasma is described as a fluid, while the REs are treated kinetically.

2.1.3 Tokamak disruptions and runaway electron generation

The major challenge of the tokamak concept is that the plasmas confined inside are susceptible to instabilities that can rapidly escalate into disruptions—events during which the plasma becomes unstable and loses its confinement very quickly (within a few milliseconds). Disruptions can cause high levels of heat loads and mechanical stress on the machine, but can also lead to the generation of REs. These electrons pose a significant threat to the PFCs [4], which are the structures that constitute the interior of the reactor and directly interact with the plasma. Because of their extreme energies and velocities, REs can deposit substantial heat loads onto PFCs, both on the surface or even deeper inside the volume, potentially causing severe damage, including melting or vaporization of wall materials. For these reasons, avoidance of disruption events wherever possible is essential, and efficient disruption mitigation is needed as a second line of defense in case of unexpected plasma dynamics or system failure, to prove the viability of the tokamak as a fusion reactor. A summary of the key physical aspects of disruptions and an overview of the different mechanisms behind the generation of REs is given here.

Tokamak disruptions

The origin of a disruption can differ widely, but the disruption dynamics generally follow the same stages. First, the temperature of the plasma decreases quickly over a short time scale (typically ≤ 1 ms), this step is called the thermal quench (TQ)[24]. The TQ is usually triggered through plasma instabilities resulting from MHD activity such as tearing modes or by an influx of impurities in the plasma which cause the cooling of the plasma through radiation and heat transport. In such a phase, the nested flux surfaces of the plasma equilibrium are broken into a stochastic structure where field lines from the interior of the plasma are directly connected to the wall structures, allowing electrons to quickly transport heat along the field lines out of the plasma core. As soon as the temperature has dropped to the range of ≈ 100 eV, this parallel heat transport becomes less efficient, and radiation by background impurities becomes the dominant loss mechanism, further dropping the temperature of the plasma into the range of 1-10 eV. This low temperature causes collisions between particles to increase, which in turns leads to an increased plasma resistivity and thus limits the ability of the plasma to carry a toroidal current. As a consequence, the ohmic plasma current carried by thermal (low energetic) electrons starts to decay.

The second step of a disruption is called the current quench (CQ), during which the plasma current decays, on a time scale that typically is much longer than that of the TQ (typically tens to hundreds of ms). According to Faraday's law, the decrease of the plasma current implies that an electric field is induced which is proportional to the current decay rate. If the decay is fast enough, the induced electric field can reach values high enough for RE generation. During the CQ phase, the fast changes to the plasma equilibrium additionally lead to a loss of the vertical control of the plasma, which is normally vertically elongated for good plasma performance and needs active position control by poloidal field coils located above and below the torus. This loss of control can lead to a vertical displacement event (VDE) which is when the plasma moves vertically into the wall of the machine. In future high current tokamaks such as ITER, RE generation will occur fast enough to create a large RE current prior to the loss of the plasma during vertical motion. In such a situation, when a large enough fraction of the current present before the disruption gets converted into current carried by the REs, the disruption will transition into the RE plateau. During the plateau phase the RE current has reached a stable level

and will make up most of the current present in the tokamak. Consequently, the current decay stops since the established RE current undergoes far less collisions and therefore remains nearly constant in time.

A fourth phase can also sometimes be observed. It can happen that an MHD instability develops during the plateau phase leading to the expulsion of the REs from the plasma [25, 26]. These events are referred to as RE beam termination events and during these a near complete loss of the RE population can be observed within less than a millisecond. After the termination, the RE current has been re-converted into an Ohmic current carried by the cold background plasma and consequently decays similarly as in the CQ. Such a termination can cause highly localized heat deposition from the REs onto the PFCs, which can lead to melting or even explosion of the wall material. This has been observed on several tokamaks[4]. Figure 2.4 summarizes these phases together with the general trend of the evolution of the different quantities. There are approaches to turning RE termination events into a more benign form, which has successfully been explored on existing tokamaks [25, 26, 27]. The key ingredients to it have been found in the injection of a lot of material into the formed RE beam, triggering a particularly beneficial burst of MHD modes that broadly distributes the REs over the PFCs without local hotspots. However, the extrapolation to large future tokamaks is still very uncertain. Predictive simulations of RE terminations have already been carried out [28, 29, 30], however without exact knowledge of the RE phase-space distribution, large uncertainties remain.

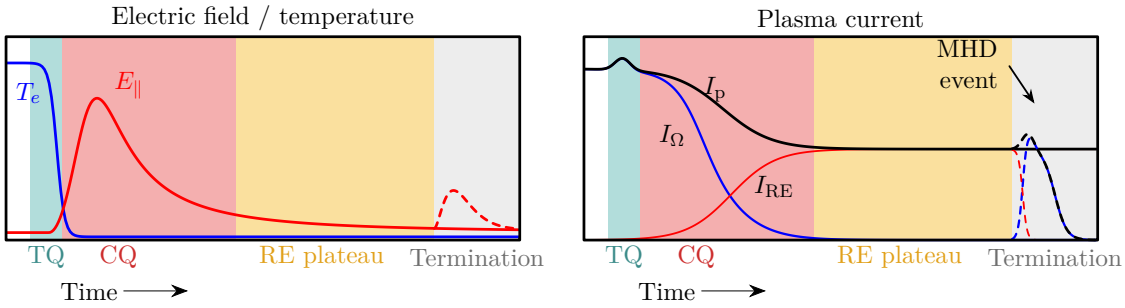


Figure 2.4: Illustration of the main stages of a disruption. The thermal quench (TQ) ends with the collapse of the electron temperature, while the current quench (CQ) ends once the Ohmic current (I_Ω) has decayed away. What remains is a runaway current (I_{RE}) that sustains the plasma current throughout the RE plateau. An MHD instability arising during this stage can be violent enough to expel the runaway population on a very short timescale, terminating the beam (dashed lines).

It is also worth mentioning that there exist several strategies to mitigate disruptions and their consequences. The most studied one is called massive material injection and it consists of injecting large amounts of material into the plasma to cool it down by increasing the radiation and therefore avoiding large heat loads on the PFCs [31]. This strategy can be achieved in different ways but the most common ones are through massive gas injection and through injection of cryogenic pellets called shattered pellet injection. To specifically mitigate runaway electrons there also exist a few strategies which are currently being explored. One possibility is the use of an external coil which aims to perturb the magnetic field in order to increase the radial transport of the REs [32]. Another well researched strategy is the benign termination scenario already described above.

Runaway electron generation

For runaways to be generated, the electric field needs to get above a certain threshold value, referred to as the Connor-Hastie electric field E_c[33]. Above this value, the electrons overcome the dragforce and can get accelerated to near lightspeed velocities. It is given by:

$$E_c = \frac{e^3 n_e \ln \Lambda}{4\pi \epsilon_0^2 m_e^2 c^2}, \tag{2.1.10}$$

where e is the electron charge, n_e is the electron density, m_e is the electron mass, ϵ_0 is the vacuum permittivity and $\ln \Lambda$ is the so-called Coulomb logarithm which is related to the Coulomb collisions the particles observe.

The generation of runaway electrons happens through different mechanisms usually classified into two categories: primary and secondary, where the distinction is made if the runaways are generated from an existing runaway population or not [3].

The primary generation includes the Dreicer generation, the hot-tail mechanism, the tritium-beta decay and the Compton scattering generation. Dreicer generation is the term used for electrons "sliding away" into the runaway region when bypassing the collisional forces. This mechanism is thus very dependent on the electric field strength.

The hot-tail mechanism happens when the plasma cools down rapidly enough so that electrons with high enough velocities (i.e., the ones in the hot tail of the Maxwellian distribution) do not have the time to slow down, for example during the TQ phase. When this happens, the high energetic REs of the Maxwellian distribution separate in phase-space from the main electron population that has cooled down due to collisions. As a consequence, if the electric field is rapidly increased, as happens during the CQ phase, the electrons contained in the hot-tail will become REs. Two other common primary mechanisms are the so-called "nuclear sources": Tritium decay and Compton generation. These will not be detailed here as they are not used in the work of this thesis.

The most important secondary generation mechanism is the so-called avalanche generation. This occurs when there is a large enough momentum exchange during a Coulomb collision between a RE and another electron (large angle collision). This collision may result in both electrons having high enough velocities to be situated in the RE regime. This process is proportional to the number of existing runaways, it therefore leads to an exponential increase of the RE population. The multiplication of a single initial RE by this process depends exponentially on the plasma current and can exceed a factor of 10^{10} in a device like ITER [3].

2.2 The JOREK non-linear extended MHD code

Transient events in tokamaks often occur on a combination of longer and shorter timescales, while also involving plasma dynamics on large and small spatial scales. Disruptions are an example for such a spatial and temporal multi-scale problem, in which the CQ happens over a particularly long time, hundreda of milliseconds in future large machines, while the gyromotion of the runaway electrons is on the order of nanoseconds. Furthermore, these processes happen on entirely different physical scales: the disruption is a macroscopic phenomenon whereas the particle motion is on the microscopic scale. To simulate both of these effects simultaneously, while treating the plasma and the RE evolution with high-fidelity, the computational power required is significant and one needs to use high-performance computing codes. The JOREK code is one of those, and it will be the focus of this section.

JOREK is a widely utilized 3D nonlinear, extended MHD and hybrid fluid-kinetic code used to study a wide range of fusion related topics [14]. It was first developed in the 2000s at CEA IRFM to study so-called edge localized modes and MHD instabilities in tokamak plasmas but has been extended over time and can today investigate large scale transient plasma dynamics in realistic tokamak geometries, notably disruption studies. The spatial discretization is done using a 2D Finite Element Method (FEM) coupled with a spectral approach in toroidal direction, while the temporal discretization is fully implicit. The code is primarily written in Fortran 90/95 with some subroutines written in C/C++ and it is parallelized using a hybrid MPI + OpenMP approach. In Subsection 2.2.1, the MHD model used in JOREK along with the model used for the tracking of test particle dynamics will be introduced. Then Subsection 2.2.2 presents the numerical discretization used in the JOREK code.

2.2.1 Physical models in JOREK

JOREK models the macroscopic plasma dynamics with an MHD framework, which is described first. For the runaway electrons, two kinetic approaches are available. The first is a 3D hybrid fluid-kinetic model [23], in which the REs are coupled self-consistently to the background plasma; it is not detailed here. The second, presented in this work, is a test particle approach, in which the REs are pushed in the fields computed by the MHD model without acting back on the background plasma, hence the name "test particle".

The JOREK reduced MHD model

To describe the statics and dynamics of a plasma, several models are possible. As mentioned earlier, the kinetic model is one of those. Another one is the MHD model, which treats the plasma as a fluid and allows to describe its macroscopic behaviour. The MHD model is formulated as a set of normalized equations combining Maxwell's equations for electromagnetism with the fluid equations resulting from taking moments of the Boltzmann equation as previously mentioned. The derivation of the MHD system is left out for simplicity, but is for example detailed in [21, 20]. However, one thing to keep in mind is that when deriving the MHD equations, the plasma is separated into two distinct populations: the electron and the ion populations, which is the so-called two-fluid model. With this model, the fluid equations are derived from the Boltzmann equation for each species. However, these two species are then combined into one, which leads to the plasma being treated as a single fluid.

Several MHD models are available in JOREK [14], but in this work a simplified MHD model is used called the reduced MHD model. In this thesis, the influence of the RE population on the general plasma dynamics modeled by the MHD system is studied. To include these effects, the current density carried by the runaway electrons $\boldsymbol{J_r}$ is subtracted from the total plasma current density $\boldsymbol{J}$ in Ohm's law (2.2.4) to correctly capture the resistive decay of the electric field in JOREK, while the runaway current is evolving. This strategy is referred to as the current coupling scheme. The set of MHD equations used in this work is then the following:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{V}) = \nabla \cdot (\underline{\mathbf{D}} \nabla \rho) \tag{2.2.1}$$

$$\rho \left(\frac{\partial \mathbf{V}}{\partial t} + \mathbf{V} \cdot \nabla \mathbf{V} \right) = -\nabla p + \mathbf{J} \times \mathbf{B} + \nabla \cdot \underline{\boldsymbol{\tau}} \tag{2.2.2}$$

$$\frac{\partial p}{\partial t} + \mathbf{V}\cdot\nabla p = (\gamma_a - 1)\eta||\mathbf{J}||^2 - \gamma_a p\nabla\cdot\mathbf{V} + (\gamma_a - 1)\nabla\cdot(\underline{\boldsymbol{\kappa}}\nabla T) + (\gamma_a - 1)\underline{\boldsymbol{\tau}} : \nabla\mathbf{V} \tag{2.2.3}$$

$$\mathbf{E} = -\mathbf{V}\times\mathbf{B} + \eta(\mathbf{J} - \mathbf{J}_r) \tag{2.2.4}$$

$$\nabla\times\mathbf{E} = -\frac{\partial\mathbf{B}}{\partial t} \tag{2.2.5}$$

$$\nabla\times\mathbf{B} = \mu_0\mathbf{J} \tag{2.2.6}$$

$$\nabla\cdot\mathbf{B} = 0. \tag{2.2.7}$$

Equations (2.2.1), (2.2.2) and (2.2.3) are the zeroth, first and second order moments of the Boltzmann equation and are referred to as the continuity equation, the momentum equation and the pressure equation respectively. Equation (2.2.4) is called Ohm's law and is derived from the momentum equation for the electrons through certain simplifications. This equation is needed to determine the evolution of the plasma current. Moreover, equations (2.2.5), (2.2.6) and (2.2.7) are the Faraday, Ampère and Gauss' law from the Maxwell system of equations.

In these equations, ρ is the mass density of the plasma, $\mathbf{V}$ is the mean velocity of the plasma, and p is the thermal pressure. The magnetic field is denoted $\mathbf{B}$, while $\mathbf{E}$ is the electric field, related to $\mathbf{B}$ through Ampère's law (with μ_0 the permeability of free space) and to $\mathbf{E}$ through Faraday's law of induction. The plasma resistivity η governs Ohmic dissipation of the magnetic field, appearing as a resistive electric field term $\eta(\mathbf{J} - \mathbf{J}_r)$ and as Ohmic heating $\eta\|\mathbf{J}\|^2$ in the pressure equation. The momentum equation includes the Lorentz force $\mathbf{J}\times\mathbf{B}$, the pressure gradient force $-\nabla p$, and the divergence of the viscous stress tensor $\underline{\boldsymbol{\tau}}$, which also drives viscous heating through $\underline{\boldsymbol{\tau}} : \nabla\mathbf{V}$ in the energy equation. The adiabatic index γ_a relates pressure changes to compressive heating/cooling via $-\gamma_a p\nabla\cdot\mathbf{V}$ and scales the remaining source terms in the pressure equation. Diffusive transport is captured by two tensor coefficients: $\underline{\mathbf{D}}$, the mass diffusivity tensor in the continuity equation, and $\underline{\boldsymbol{\kappa}}$, the thermal conductivity tensor, which drives heat conduction via $\nabla\cdot(\underline{\boldsymbol{\kappa}}\nabla T)$, with T the plasma temperature. Finally, ∇, $\nabla\cdot$, and $\nabla\times$ denote the gradient, divergence, and curl operators, and $\partial/\partial t$ the partial time derivative, while $\nabla\cdot\mathbf{B} = 0$ is the solenoidal constraint expressing the absence of magnetic monopoles.

It is worth mentioning that the MHD equations allow for the development of a wide range of instabilities. These can be broadly classified according to their toroidal mode number n, i.e. the number of times a perturbation repeats around the torus toroidally: axisymmetric instabilities ($n = 0$) include for example the VDE, while non-axisymmetric instabilities ($n > 0$) include for example the previously mentioned tearing mode. Since JOREK-DREAM aims at intermediate fidelity simulations excluding 3D plasma dynamics, only $n = 0$ instabilities are considered throughout this thesis, the most relevant of which is the VDE, as it is the main instability observed during the CQ and plateau phases (Section 2.1.3).

The previous MHD equations are coupled to the following reduced MHD ansatz:

$$\mathbf{B} = \nabla\psi\times\nabla\varphi + F_0\nabla\varphi, \tag{2.2.8}$$

where $F_0 = B_\varphi R$ is a spatial and temporal constant. The perpendicular flows are modeled through the velocity stream function $v = \phi/F_0$, where ϕ is the electric potential and the temperature and the pressure are related through the ideal gas law $p = \rho T_{\mathrm{MHD}}$, where the pressure is defined by $p = p_e + p_i$ and the "MHD" temperature by $T_{\mathrm{MHD}} = T_e + T_i$. The distinction made for the pressure and the temperature comes from the two-fluid model mentioned above. This then leads to the following system of simplified equations [14, 34]:

$$\frac{\partial \psi}{\partial t} = R[\psi, v] + \eta(j - j_r) - F_0 \frac{\partial v}{\partial \varphi} - R\eta_{\text{num}} \nabla^2 \left(\frac{j}{R} \right) \tag{2.2.9}$$

$$\frac{\partial \rho}{\partial t} = R[\rho, v] + 2\rho \frac{\partial v}{\partial Z} + \nabla \cdot \left(D_\parallel \nabla_\parallel \rho + D_\perp \nabla_\perp \rho \right) \tag{2.2.10}$$

$$\begin{aligned} \nabla \cdot \left[R^2 \rho \nabla_{pol} \left(\frac{\partial v}{\partial t} \right) \right] &= \frac{1}{R} \left[R^4 \rho \omega, u \right] - \frac{1}{2R} \left[R^2 \rho, R^2 |\nabla_{pol} v|^2 \right] \\ &\quad - \frac{1}{R} \left[R^2, p \right] + \frac{1}{R} \left[\psi, j \right] - \frac{F_0}{R^2} \frac{\partial j}{\partial \varphi} + \mu \nabla^2 \omega \end{aligned} \tag{2.2.11}$$

$$\begin{aligned} \frac{\partial p}{\partial t} &= R[p, v] + 2\gamma_a \rho T_{\text{MHD}} \frac{\partial v}{\partial Z} + \nabla \cdot \left(\chi_\parallel \nabla_\parallel T_{\text{MHD}} + \chi_\perp \nabla_\perp T_{\text{MHD}} \right) \\ &\quad + (\gamma_a - 1) \frac{\eta}{R^2} j^2 \end{aligned} \tag{2.2.12}$$

$$j = \Delta^* \psi \tag{2.2.13}$$

$$\omega = \Delta_{pol} v. \tag{2.2.14}$$

In the above equations, $[a, b] = \partial_R a \, \partial_Z b - \partial_Z a \, \partial_R b$ represents the Poisson bracket in the (R, Z) coordinate system, $j = J_\varphi / R$ is the normalized toroidal current density, j_r is the normalized toroidal runaway current density and ω is the vorticity. Δ_{pol} and ∇_{pol} denote the Laplacian and the gradient in the (R, Z) coordinate system. Furthermore, $D_\parallel$, $D_\perp$, $\chi_\parallel$ and $\chi_\perp$ denote the parallel and perpendicular particle and heat transport coefficients respectively. These are defined as input parameters to the simulations. μ denotes the viscosity. Equations (2.2.9) - (2.2.14) are the equations solved in JOREK in order to determine the unknown quantities which are the poloidal magnetic flux ψ, the mass density ρ, the poloidal flow potential v, the thermal pressure p, the toroidal current density j, and the toroidal vorticity ω. Equations (2.2.9)–(2.2.14) are defined on the poloidal cross-section $\Omega \subset (R, Z)$-space, which represents a poloidal plane of the tokamak (bounded by a specified computational boundary), together with time $t \in [0, T]$. The unknowns ψ, ρ, v, p, j, and ω are scalar fields

$$\psi, \rho, v, p, j, \omega : \Omega \times [0, T] \to \mathbb{R}, \tag{2.2.15}$$

each mapping a point (R, Z, t) in the poloidal domain and time interval to a real value.

In addition, JOREK has the capability to self-consistently evolve the free plasma boundary and the resistive wall currents through its coupling to the STARWALL [35] or the CARIDDI [36] code, which is essential for the simulation of VDEs.

Model for test particles in JOREK

As explained, the aim of the JOREK-DREAM coupling is to not simulate the particles using the 3D hybrid fluid-kinetic approach. However, for later purposes we will need to evolve certain particles kinetically which we will simulate using the test particle motion capability of the code, which is detailed here. These particles are modeled kinetically using a Monte Carlo, also called particle-in-cell (PIC), approach, where a population is represented by a discrete set of markers called test particles. Instead of solving the Boltzmann equation (2.1.5) to determine the continuous evolution of the distribution function, a statistical sampling of the distribution function is performed, giving the following description of the discretized distribution function:

$$f(\boldsymbol{x}, \boldsymbol{v}, t) = \sum_{i=1}^{N_m} w_i \delta(\boldsymbol{x} - \boldsymbol{x}_i(t)) \delta(\boldsymbol{v} - \boldsymbol{v}_i(t)), \tag{2.2.16}$$

where N_m is the total number of simulated markers, w_i the weight associated to marker i, which represents the number of physical particles it contains and where $\boldsymbol{x}_i(t)$ and $\boldsymbol{v}_i(t)$ are the position and velocity of marker i at time t. There are two kinds of RE test particles in JOREK [37]: one resolves the full gyro orbit of the particle (FO marker) and the second uses a gyrokinetic (guiding center) approximation (GC marker). The GC markers are computationally cheaper to simulate than the full orbit ones. The model and implementation of the full orbit markers will not be detailed here but it can be found in [37]. The relativistic guiding center (GC) model used in JOREK is described by [38, 39]:

$$\dot{\mathbf{X}} = \frac{1}{\mathbf{b}\cdot\mathbf{B}^*}\left(q\mathbf{E}\times\mathbf{b} - p_\parallel\frac{\partial\mathbf{b}}{\partial t}\times\mathbf{b} + \frac{m\mu\mathbf{b}\times\nabla B + p_\parallel\mathbf{B}^*}{m\gamma_{\mathrm{GC}}}\right) \tag{2.2.17}$$

$$\dot{p}_\parallel = \frac{\mathbf{B}^*}{\mathbf{b}\cdot\mathbf{B}^*}\cdot\left(q\mathbf{E} - p_\parallel\frac{\partial\mathbf{b}}{\partial t} - \frac{\mu\nabla B}{\gamma_{\mathrm{GC}}}\right) \tag{2.2.18}$$

$$\gamma_{\mathrm{GC}} = \sqrt{1 + \left(\frac{p_\parallel}{mc}\right)^2 + \frac{2\mu B}{mc^2}} \tag{2.2.19}$$

where $\mathbf{X}$ is the GC position vector, $p_\parallel$ is the particle momentum in the direction of the magnetic field, $\mu = \frac{\|\mathbf{p}-p_\parallel\mathbf{b}\|^2}{2mB}$ is the magnetic moment, B is the magnetic field magnitude, $\mathbf{b} = \frac{\mathbf{B}}{B}$ the magnetic field direction and $\mathbf{B}^* = p_\parallel\nabla\times\mathbf{b} + q\mathbf{B}$ is the so-called "effective magnetic field".

The guiding center markers are evolved using the fifth order Cash-Karp Runge-Kutta scheme described in [40], which is not described here due to its complexity. The scheme is explicit, which means that energy is not exactly conserved, but with sufficiently small time steps the error can be made small enough [37].

2.2.2 Numerical methods employed in JOREK

The JOREK code uses the cylindrical coordinate system (R, Z, φ) as shown in Figure 2.2. The fields are discretized using G^1-continuous isoparametric 2D Bézier finite elements in the poloidal cross-section of the simulation domain, combined with a Fourier discretization in the toroidal direction [41]. The Fourier decomposition will not be detailed here as no $n > 0$ dynamics are considered, as mentioned in Subsection 2.2.1. Within each element, a local coordinate system is defined using local coordinates s and t. In this work, bi-cubic elements (polynomial degree $m = 3$, using 16 basis functions per element) are used throughout, following the standard choice in JOREK; higher-order elements (e.g. bi-quintic) are supported by the code but not considered here [42]. The same set of basis functions is used both to represent the physical domain (geometry) and to approximate the physical variables, i.e. an isoparametric representation. The description given here follows the works of [41] and [42].

Bézier elements

For this finite element method, Bernstein polynomials of degree m are used along each of the two local coordinates $s, t \in [0, 1]$, defined by:

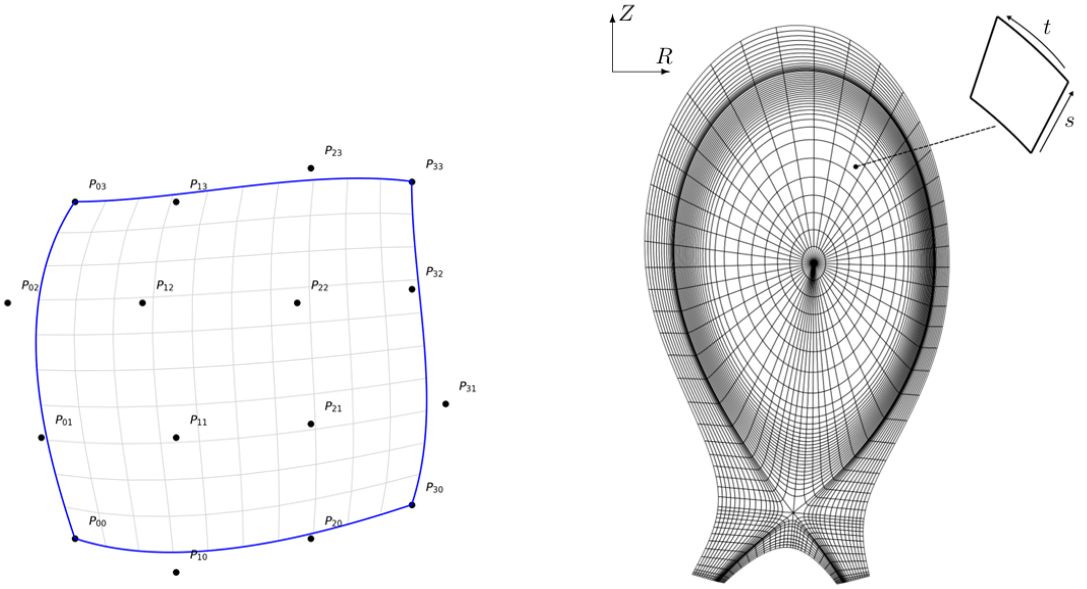
$$B_i^m(s) = \frac{m!}{i!(m-i)!}s^i(1-s)^{m-i} \quad \text{for } i = 0, ..., m. \tag{2.2.20}$$

These polynomials satisfy a few very useful properties: they constitute a basis of Π_m (the subspace of polynomials whose degree is less than or equal to m), they are non-negative, and they form a partition of unity for $s \in [0, 1]$.

A Bézier surface is then constructed from a grid of control points $(P_{i,j}) \in \mathbb{R}^d$, where d is the dimension of the space. A representation of a bi-cubic Bézier surface with its control points is given in Figure 2.5a. A bi-cubic element is a finite element in which the local geometry and physical fields are represented using polynomials of degree three (cubic) independently in each of the two local coordinate directions (s) and t), giving 16 basis functions per element (4 per direction, combined via a tensor product). A given field $F(s,t)$ on a 2D Bézier element is expressed as:

$$F(s,t) = \sum_{i=0}^{3} \sum_{j=0}^{3} B_i^3(s) B_j^3(t) P_{i,j}. \tag{2.2.21}$$

Taking $d = 2$ and $F = (R, Z)$ in this same expression gives the real-space element geometry itself – the isoparametric property mentioned above. This allows the elements to be shaped such that they align with the flux surfaces of the plasma as illustrated in Figure 2.5b. It is possible to align the grid to the magnetic geometry since this allows to tune the spatial resolution to better capture dynamics parallel and perpendicular to the magnetic field lines separately.



(a) (b)

Figure 2.5: (a) Bi-cubic Bézier surface with its associated control points. Reprint from [43]. (b) Illustration of a flux surface aligned finite element grid in JOREK including a realistic x-point geometry. A single Bézier element is also represented with its local coordinates (s,t). Reprint from [34].

Since neighbouring elements are represented independently, care must be taken to join them smoothly. Consider two elements sharing an edge, and let P_0 denote a shared control point on that edge, with P_A and P_B the neighbouring control points on either side, belonging respectively to the two elements. Geometrical continuity is classified in a hierarchy of increasing smoothness, defined as follows:

$$G^0 \text{ continuity:} \quad P_0^{(1)} = P_0^{(2)}, \tag{2.2.22}$$

$$G^1 \text{ continuity:} \quad P_A - P_0 = -\alpha \left(P_B - P_0 \right), \quad \alpha > 0, \tag{2.2.23}$$

$$C^1 \text{ continuity:} \quad P_A - P_0 = -\left(P_B - P_0\right). \tag{2.2.24}$$

G^0 continuity simply requires adjacent elements to share the same control point along their common edge (Equation (2.2.22)). G^1 continuity additionally requires the tangent vectors $P_A - P_0$ and $P_B - P_0$ to be collinear across the edge (Equation (2.2.23)), but allows their magnitudes to differ by a scalar factor α. C^1 continuity is the special case $\alpha = 1$ (Equation (2.2.24)), where the tangent vectors must additionally match in magnitude. G^1 continuity is therefore strictly less restrictive than C^1 continuity, since it constrains only the direction and not the amplitude of the tangent vectors. Imposing G^2 continuity further constrains the curvature to match across the boundary, which significantly reduces the flexibility of the element shapes unless higher-order (e.g. bi-quintic) elements are used [42].

JOREK typically uses G^1 continuity in production applications as a compromise: it is smooth enough to support the second-order spatial derivatives appearing in the diffusive terms of the reduced MHD system (Eqs. (2.2.1)-(2.2.3)), which can be recast into first-order form via integration by parts or auxiliary variables, while remaining more flexible for mesh generation than the fully C^1-continuous alternative.

Nodal representation

Naively, each bi-cubic element as introduced above carries 16 independent control points $P_{i,j}$, $i, j = 0, \ldots, 3$, and hence 16 degrees of freedom. However, this counts every control point as if it belonged uniquely to a single element, which is inconsistent with the continuity conditions of Equations (2.2.22),(2.2.23), which introduce constraints between the control points of neighbouring elements that must be reflected in how the degrees of freedom are counted and assembled globally.

To handle this consistently, JOREK adopts a node-based representation instead of an element-based one [41, 42]. Consider a node shared by four neighbouring elements, for example $P_{0,0}$, illustrated in Figure 2.6. The three control points inside a given element that are adjacent to this vertex are then defined from $P_{0,0}$ using a set of nodal vectors $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$:

$$P_{1,0} = P_{0,0} + h^u \boldsymbol{u}, \tag{2.2.25}$$
$$P_{0,1} = P_{0,0} + h^v \boldsymbol{v}, \tag{2.2.26}$$
$$P_{1,1} = P_{0,0} + h^u \boldsymbol{u} + h^v \boldsymbol{v} + h^w \boldsymbol{w}, \tag{2.2.27}$$

where the coefficients h^u, h^v and h^w, referred to as element sizes, set the distance between the vertex and its neighbouring control points along $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$. The vectors $\boldsymbol{u}$, $\boldsymbol{v}$, $\boldsymbol{w}$ are proportional to the s-, t- and mixed s-t derivatives of the represented quantity at the node [42]. For the geometry, $P_{0,0}$ and the nodal vectors are vectors in the (R, Z) plane, while for a physical field they are scalars.

The nodal quantities $P_{0,0}$, $\boldsymbol{u}$, $\boldsymbol{v}$, $\boldsymbol{w}$ are the unknowns of the problem and are shared by the four elements around the node, whereas the element sizes may differ from element to element. The element sizes are fixed by the grid and constrained such that continuity holds for any value of the nodal unknowns: G^0 continuity requires equal sizes on both sides of a shared edge, G^1 continuity requires the sizes of opposite elements to be related through the factor $\alpha > 0$ of Equation (2.2.23), e.g. $h^u_{-1,1} = -\alpha h^u_{1,1}$, where the subscripts label the four elements around the node, and the mixed term is fixed by $h^w = h^u h^v$ [42]. Continuity is therefore not imposed in the linear system, but ensured by the construction of the elements.

This is what reduces the number of degrees of freedom. Around a node shared by four elements, each element contributes four control points adjacent to that node, giving 16 in total. Imposing G^0 continuity makes the control points on the shared edges coincide, leaving 9, and imposing G^1 continuity expresses all of them through the four nodal quantities $P_{0,0}$, $\boldsymbol{u}$, $\boldsymbol{v}$, $\boldsymbol{w}$. Each element is thus still described by 16 coefficients (4 per vertex), but these are shared with the neighbouring elements, so that globally only 4 degrees of freedom per node remain.

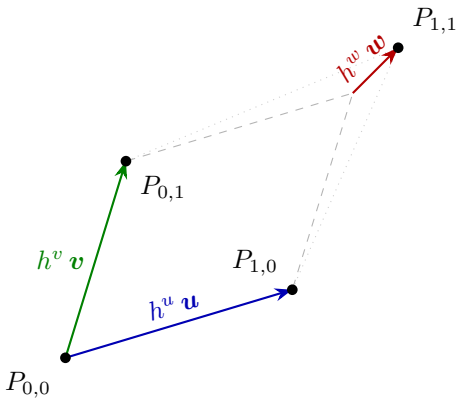


Figure 2.6: Construction of the control points $P_{0,1}$, $P_{1,0}$, and $P_{1,1}$ neighbouring a shared vertex $P_{0,0}$, from the local basis vectors $\boldsymbol{u}$, $\boldsymbol{v}$, $\boldsymbol{w}$ and coefficients h^u, h^v, h^w.

Using the node-based representation, a physical field G on a given element is represented as

$$G(s,t) = \sum_{i=1}^{4}\sum_{j=1}^{4} G_{i,j}\, h_{i,j}\, H_{i,j}(s,t), \tag{2.2.28}$$

where $H_{i,j}(s,t)$ are the basis functions of the nodal representation, $G_{i,j}$ is the coefficient corresponding to the j-th nodal degree of freedom of the i-th vertex, and $h_{i,j} \in \{1, h_i^u, h_i^v, h_i^u h_i^v\}$ is the corresponding element size.

Temporal discretization

Having established the spatial discretization of the fields, attention is now turned to the temporal discretization of the fluid equations (2.2.9)-(2.2.14). The discretization of the fluid equations in JOREK is fully implicit in time, which allows the code to take much larger time steps than an explicit scheme would permit, at the cost of having to solve a large linear system at every time step. The equations are solved in weak form, obtained by multiplying each equation by a test function, integrating over the simulation domain, and transferring derivatives onto the test function through integration by parts where necessary; the resulting integrals are evaluated with Gaussian quadrature. To describe this, the weak forms of the fluid equations are first written in the generic operator form:

$$\frac{\partial A(\boldsymbol{u},t)}{\partial t} = B(\boldsymbol{u},t), \tag{2.2.29}$$

where $\boldsymbol{u} = (\psi, \rho, v, p, j, \omega)$ is the vector containing all evolved physical variables, and A and B are integro-differential operators encapsulating the weak form of Equations (2.2.9)-(2.2.14).

Let t_n be the time at which a quantity is evaluated, and let $\delta \boldsymbol{u}^n = \boldsymbol{u}^{n+1} - \boldsymbol{u}^n$, where $\boldsymbol{u}^n \approx \boldsymbol{u}(t_n)$. Since the operators A and B are nonlinear, they are linearized around the current time step using a first-order Taylor expansion,

$$A(\boldsymbol{u}^{n+1}) \approx A(\boldsymbol{u}^n) + \left(\frac{\partial A}{\partial \boldsymbol{u}}\right)^n \cdot \delta \boldsymbol{u}^n, \tag{2.2.30}$$

$$B(\boldsymbol{u}^{n+1}) \approx B(\boldsymbol{u}^n) + \left(\frac{\partial B}{\partial \boldsymbol{u}}\right)^n \cdot \delta \boldsymbol{u}^n. \tag{2.2.31}$$

An analogous backward linearization is used to express quantities at the previous time step $n-1$ in terms of quantities at time step n,

$$A(\boldsymbol{u}^{n-1}) \approx A(\boldsymbol{u}^n) - \left(\frac{\partial A}{\partial \boldsymbol{u}}\right)^n \cdot \delta \boldsymbol{u}^{n-1}, \tag{2.2.32}$$

$$B(\boldsymbol{u}^{n-1}) \approx B(\boldsymbol{u}^n) - \left(\frac{\partial B}{\partial \boldsymbol{u}}\right)^n \cdot \delta \boldsymbol{u}^{n-1}. \tag{2.2.33}$$

Combining these expansions, a time discretized form of Equation (2.2.29) for a time step size Δt can be written compactly as

$$\left[(1+\zeta)\left(\frac{\partial A}{\partial \boldsymbol{u}}\right)^n - \Delta t\, \lambda \left(\frac{\partial B}{\partial \boldsymbol{u}}\right)^n\right] \cdot \delta \boldsymbol{u}^n = \Delta t\, B(\boldsymbol{u}^n) + \zeta \left(\frac{\partial A}{\partial \boldsymbol{u}}\right)^n \cdot \delta \boldsymbol{u}^{n-1}, \tag{2.2.34}$$

where ζ and λ are free parameters selecting the particular time integration scheme used. Setting $\zeta = 0$, $\lambda = 1/2$ recovers the Crank-Nicolson scheme, which is second-order accurate and has excellent conservation properties, while setting $\zeta = 1/2$, $\lambda = 1$ recovers the Gear scheme, which is more dissipative but better suited to damping spurious numerical oscillations. Both schemes are available in JOREK, and the choice between them is made depending on the application.

Equation (2.2.34) is still expressed in terms of the continuous operators A and B, acting on the unknown $\delta \boldsymbol{u}^n$ as a function of space; solving it in practice requires spatially discretizing both sides using the spatial disretization and nodal representation introduced above. After this is done, a linear system is obtained:

$$C^{\text{var}}(\delta \mathbf{u}^n)^{\text{nodal}} = \mathbf{d}^{\text{var}}, \tag{2.2.35}$$

where the mass matrix C^{var} represents the spatial discretization in nodal form of the left-hand side matrix of Equation (2.2.34) and the load vector $\mathbf{d}^{\text{var}}$ the spatial discretization in nodal form of the right-hand side vector of Equation (2.2.34). The superscript var denotes that the matrix C^{var} is assembled by blocks where each block represents the coupling of one variable with another and similarly the vector $\mathbf{d}^{\text{var}}$ concatenates the nodal representations for each variable. Furthermore, $(\delta \mathbf{u}^n)^{\text{nodal}}$ is the vector of nodal coefficients of the increment $\delta \mathbf{u}^n$.

Since only axisymmetric ($n = 0$) dynamics are considered in this work, this sparse linear system is solved at every time step with a direct solver, using one of the libraries PaStiX [44], MUMPS [45] or STRUMPACK [46].

2.3 The DREAM bounce averaged fluid-kinetic code

The computational cost of hybrid fluid-kinetic RE simulations [23] in JOREK becomes unjustified over periods where 3D MHD activity does not play a major role. In these scenarios, the model can be greatly simplified by exploiting the toroidal and approximate

poloidal symmetry of the tokamak. If the drift of the particles is also neglected and if it is assumed that these follow the magnetic field lines exactly (the zero drift-orbit limit), the electron dynamics over the fast gyromotion and the bounce motion along a field line can be averaged, retaining only the slow evolution in the remaining phase-space coordinates. These are the assumptions employed in the DREAM code [16].

The DREAM code was developed in the early 2020s by the Plasma Theory group at the Chalmers University of Technology. The code was first developed to study the dynamics of runaway electrons in tokamak disruptions. It has been extended over time to include more models. The spatial discretization is done using a 1D Finite Volume Method (FVM) and the temporal discretization is fully implicit. The core of the code is written in C++ and it is run via a Python interface and parallelization is on the way. In Subsection 2.3.1 the physical models used in DREAM are described. Then in Subsection 2.3.2 the numerical methods used to discretize the governing equations are detailed.

2.3.1 Physical models in DREAM

In DREAM, the background plasma parameters, such as the electron temperature, electron and ion densities and the electric field can be evolved self-consistently, and are coupled to the electron kinetics through the collision operator and Ohm's law. In this work, many of these background quantities will instead be prescribed based on the JOREK output, as detailed in Chapter 3. Four models of increasing complexity are available in DREAM for the electron dynamics: fluid, isotropic, superthermal and fully kinetic [16]. These trade physical fidelity against computational cost, primarily through how much of momentum space is resolved kinetically rather than approximated by fluid closures. In this thesis, only the fully kinetic model is used which provides an accurate evolution of the electron distribution function over time.

Three electron populations

The electron dynamics in a post-disruption plasma are qualitatively different on three, typically well-separated momentum scales [16]: the thermal momentum $p_{Te} \sim 0.01\, m_e c$, at which the ohmic current and Joule heating are set; the critical momentum $p_c \sim \sqrt{E_c/E_\parallel} \sim 0.1\, m_e c$, at which electrons transition from being collisionally thermalized to being continuously accelerated, as described in Section 2.1.3; and the acceleration region $p \sim \int eE_\parallel \, \mathrm{d}t/m_e c \sim 100\, m_e c$, in which the energy spectrum of the runaway tail, and hence its synchrotron and bremsstrahlung emission, is set. Exploiting this separation, DREAM allows for the splitting of the electron population into three momentum regions: cold, hot and runaway, characterized by densities n_{cold}, n_{hot} and n_{re}, respectively. These three different populations are illustrated in Figure 2.7. The cold and hot electrons are, in general, described by a shared distribution function f_{hot}, occupying $0 \leq p < p_{\mathrm{re}}$, while the runaway electrons are described by a separate distribution function f_{re} defined on its own numerical grid spanning $p \geq p_{\mathrm{re}}$; the two grids are coupled through a particle-conserving boundary condition at p_{re} [16]. The values p_{hot} and p_{re} are set by the user.

The fully kinetic model

In the fully kinetic model, both the cold and hot populations are resolved by a single distribution function $f_{\mathrm{hot}}(t, r, p, \xi_0)$, evolved according to a bounce-averaged drift-kinetic equation (similar to the Fokker-Planck equation (2.1.6) introduced earlier), given for ar-

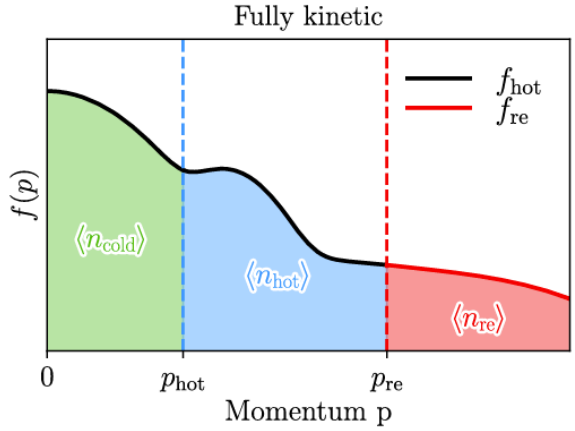


Figure 2.7: Illustration of the three electron populations in the fully kinetic model. The distribution function f_{hot} describes both the cold ($p < p_{hot}$) and hot ($p_{hot} \leq p < p_{re}$) electrons, while f_{re} describes the runaway ($p \geq p_{re}$) electrons on a separate simulation grid. Reprint from [16].

bitrary distribution function f by:

$$\frac{\partial f}{\partial t} = \sum_{m,n} \frac{1}{\mathcal{V}'} \frac{\partial}{\partial z^m} \left[\mathcal{V}' \left(-\{A^m\} f + \{D^{mn}\} \frac{\partial f}{\partial z^n} \right) \right] + \{S\}, \tag{2.3.1}$$

where $\mathbf{z} = (r, p, \xi_0)$ are constants of the (zero-orbit-width) particle motion: r is the flux-surface label, p the momentum magnitude, and ξ_0 the pitch with respect to the magnetic field evaluated at the point of minimum field strength B_{min} on the flux surface. The summing indices m and n denote the phase space coordinates. $\mathcal{V}'$ denotes the jacobian in the phase space. The curly brackets denote an average over the fast gyromotion and the bounce motion along the magnetic field line. The derivation of the bounce-averaged drift-kinetic equation (2.3.1) is detailed in [47]. The runaway population is described by a second distribution function, $f_{\mathrm{re}}(t, r, p, \xi_0)$, which satisfies the same equation (2.3.1), but is defined on a separate momentum grid spanning $p_{\mathrm{re}} \leq p < p_{\mathrm{max}}$. The two distribution functions are connected via a particle-conserving flux boundary condition at $p = p_{\mathrm{re}}$, described further in Section 2.3.2.

The advection and diffusion coefficients in Equation (2.3.1) contain the physics governing the electron dynamics: acceleration by the parallel electric field, the collision operator for electron-electron and electron-ion collisions among others. The source term $\{S\} = \{C_{\mathrm{ava}}\} + \{S_p\}$ contains the Rosenbluth-Putvinski avalanche operator,

$$\{C_{\mathrm{ava}}\} = n_{\mathrm{re}}\, n_{\mathrm{tot}}\, c\, r_0^2 \left\{ \frac{B\, \delta(\xi - \xi^\star)}{\langle B \rangle} \right\} \frac{1}{p^2} \frac{\partial}{\partial p} \frac{1}{1-\gamma}, \qquad \xi^\star = \sqrt{\frac{\gamma - 1}{\gamma + 1}}, \tag{2.3.2}$$

where n_{tot} is the total (free and bound) electron density and r_0 the classical electron radius, together with an ad-hoc particle source S_p used to model electron density changes due to ionization [16].

Dreicer generation emerges directly from the evolution of f_{hot} itself. As electrons are accelerated past the boundary momentum p_{re}, they are transferred into the runaway distribution f_{re} through a particle-conserving flux boundary condition on the momentum-

space particle flux of f_{hot} across $p = p_{\mathrm{re}}$: [16]

$$F_{\mathrm{hot/re}} = \int \mathcal{V}' \left(\{A^p\} f_{\mathrm{hot/re}} + \{D^{pp}\} \frac{\partial f_{\mathrm{hot/re}}}{\partial p} \right) \mathrm{d}\xi_0 \Bigg|_{p=p_{\mathrm{re}}} . \tag{2.3.3}$$

This boundary condition is explained in more detail in Section 2.3.2.

Because the test-particle collision operator used in DREAM neglects the field-particle part of the electron-electron collision operator, the conductivity it supports is systematically underestimated relative to the true (Spitzer-like) response [16]. The ohmic current is therefore not simply the low-momentum moment of f_{hot}, but is corrected according to

$$\frac{j_\Omega}{B} = \sigma \frac{\langle \mathbf{E} \cdot \mathbf{B} \rangle}{\langle B^2 \rangle} + \frac{\delta j_{\mathrm{corr}}}{B}, \tag{2.3.4}$$

where σ is the same Sauter-Redl conductivity [48], and the correction

$$\frac{\delta j_{\mathrm{corr}}}{B} = -\frac{2\pi e}{B_{\mathrm{min}}} \int_0^{p_{\mathrm{hot}}} \mathrm{d}p \int_{-1}^{1} \mathrm{d}\xi_0 \, p^2 \mathcal{H}(\xi_0) v \xi_0 f_{\mathrm{hot}} \; - \; \sigma_{\mathrm{kineq}} \frac{\langle \mathbf{E} \cdot \mathbf{B} \rangle}{\langle B^2 \rangle} \tag{2.3.5}$$

removes the numerical ohmic current predicted by the kinetic operator itself, replacing it with the correct conductivity response. The reduced conductivity σ_{kineq} supported by the test-particle operator was found, by matching against reference calculations for $Z_{\mathrm{eff}} \in [0, 50]$, to be well approximated by

$$\sigma_{\mathrm{kineq}} = \left(1 - \frac{1.406}{1.888 + Z_{\mathrm{eff}}} \right) \sigma_0, \tag{2.3.6}$$

where σ_0 is the collisionless (Braams-Karney [49]) limit of the Sauter-Redl formula [16] and Z_{eff} is the effective charge of the plasma.

Here $\mathcal{H}$ denotes the step function:

$$\mathcal{H}(\xi_0) = \begin{cases} 1, & |\xi_0| > \xi_T \\ 0, & |\xi_0| \leq \xi_T \end{cases}, \tag{2.3.7}$$

with $\xi_T = \sqrt{1 - B_{\mathrm{min}}/B_{\mathrm{max}}}$ denoting the trapped-passing boundary: the ξ value at which particles bounce back. The hot and runaway current densities are then defined as:

$$\frac{j_{\mathrm{hot}}}{B} = -\frac{2\pi e}{B_{\mathrm{min}}} \int_{p_{\mathrm{hot}}}^{p_{\mathrm{re}}} \mathrm{d}p \int_{-1}^{1} \mathrm{d}\xi_0 \, p^2 \mathcal{H}(\xi_0) v \xi_0 f_{\mathrm{hot}}, \tag{2.3.8}$$

and:

$$\frac{j_{\mathrm{re}}}{B} = -\frac{2\pi e}{B_{\mathrm{min}}} \int_{p_{\mathrm{re}}}^{p_{\mathrm{max}}} \mathrm{d}p \int_{-1}^{1} \mathrm{d}\xi_0 \, p^2 \mathcal{H}(\xi_0) v \xi_0 f_{\mathrm{re}}. \tag{2.3.9}$$

To summarize, the fully kinetic model evolves the drift-kinetic equation (2.3.1) for the two distribution functions f_{hot} and f_{re}, defined on separate momentum grids and coupled through a flux boundary condition (2.3.3). All other quantities of interest, the cold density n_{cold}, the hot and runaway current densities j_{hot} and j_{re} (Equations (2.3.8),(2.3.9)), and the corrected ohmic current j_Ω (Equation (2.3.4)) are obtained as moments of these two distribution functions, rather than being evolved as separate unknowns.

Setting the impurity background

To respect the quasi-neutrality of the plasma, the impurity background has to be prescribed accordingly. For simplicity, it is supposed to be composed only of Deuterium (D) with density n_D (and charge 1) and fully ionised Neon (Ne) with density n_{Ne} and with charge $Z = 10$. Quasi-neutrality requires

$$n_{\mathrm{cold}} = n_{\mathrm{D}} + Z\, n_{\mathrm{Ne}}, \tag{2.3.10}$$

while the effective charge, defined as $Z_{\mathrm{eff}} \equiv \left(\sum_i n_i Z_i^2\right)/n_{\mathrm{cold}}$ (where the index i sums over the ion species in th plasma), becomes:

$$Z_{\mathrm{eff}}\, n_{\mathrm{cold}} = n_{\mathrm{D}} + Z^2 n_{\mathrm{Ne}}. \tag{2.3.11}$$

Solving Equations (2.3.10)–(2.3.11) for n_{D} and n_{Ne} gives

$$n_{\mathrm{Ne}} = n_{\mathrm{cold}}\,\frac{Z_{\mathrm{eff}} - 1}{Z(Z-1)}, \qquad n_{\mathrm{D}} = n_{\mathrm{cold}}\,\frac{Z - Z_{\mathrm{eff}}}{Z-1}. \tag{2.3.12}$$

In the coupled model, the quantity Z_{eff} is taken directly from JOREK as detailed in Section 3.2.

2.3.2 Numerical methods employed in DREAM

Discretization of the drift-kinetic equation

DREAM employs a finite volume method to discretize the kinetic equation (2.3.1). The computational grid in each phase-space direction $z^m \in \{r, p, \xi_0\}$ is divided into cells, giving rise to two interleaved grids: the distribution grid, with integer indices (i, j, k), on which the unknown quantities (e.g. f_{hot}, f_{re}) are defined, and the flux grid, with half-integer indices such as $(i - \frac{1}{2}, j, k)$, on which particle fluxes are evaluated. Grids are specified by their edge points $z^m_{1/2}$ and $z^m_{N_m+1/2}$ together with the cell widths Δz^m_i, so that

$$z^m_{i+1/2} = z^m_{i-1/2} + \Delta z^m_i, \qquad z^m_i = \frac{z^m_{i+1/2} + z^m_{i-1/2}}{2}, \tag{2.3.13}$$

with the corresponding distribution-grid spacing $\Delta z^m_{i+1/2} = z^m_{i+1} - z^m_i$.

Equation (2.3.1) is integrated over the volume of each cell, converting the divergence term into a difference of fluxes evaluated on the two bounding cell faces of each phase-space direction. Together with an implicit (backward) Euler discretization of the time derivative, this yields the discretized form:

$$\begin{aligned}
\frac{f^{(l+1)}_{ijk} - f^{(l)}_{ijk}}{\Delta t^{(l)}} =& \frac{1}{\mathcal{V}'_{ijk}\Delta z^m_{ijk}}\Big[\mathcal{V}'_{i_m+1/2}\left\{A^{m,(l+1)}\right\}_{i_m+1/2} f^{(l+1)}_{i_m+1/2} - \\
& \mathcal{V}'_{i_m-1/2}\left\{A^{m,(l+1)}\right\}_{i_m-1/2} f^{(l+1)}_{i_m-1/2} + \\
& \mathcal{V}'_{i_m+1/2}\left\{D^{mn,(l+1)}\right\}_{i_m+1/2} \left.\frac{\partial f}{\partial z^n}\right|^{(l+1)}_{i_m+1/2} - \\
& \mathcal{V}'_{i_m-1/2}\left\{D^{mn,(l+1)}\right\}_{i_m-1/2} \left.\frac{\partial f}{\partial z^n}\right|^{(l+1)}_{i_m-1/2}\Big] + \\
& \{S\}^{(l+1)}_{ijk},
\end{aligned} \tag{2.3.14}$$

where l denotes the time-step index and the notation $i_m \pm N$ indicates that N is added to or subtracted from the index of the m'th phase-space coordinate. Since fluxes are evaluated once on each cell face and shared between neighbouring cells, this scheme is conservative in the absence of sources [16].

Discretization of diffusion terms

As seen in Equation (2.3.14), evaluating the diffusive flux $\{D^{mn}\}\,\partial f/\partial z^n$ on each cell face requires an estimate of the gradient $\partial f/\partial z^n$ at the face location. For the diagonal ($m = n$) terms, this face gradient is approximated using a central difference between the two neighbouring cell-centre values:

$$\left.\frac{\partial f}{\partial z^n}\right|_{i_{m-1/2}} = \frac{f_{i_n} - f_{i_n-1}}{\Delta z^n_{i_n-1/2}}, \tag{2.3.15}$$

which preserves the monotonicity of f. The off-diagonal terms, by contrast, require a gradient in a direction transverse to the face normal, and so cannot be evaluated directly from the two adjacent cells; instead, the required value is reconstructed via interpolation between the surrounding cell-centre values:

$$\left.\frac{\partial f}{\partial z^n}\right|_{i_{m-1/2}} = \frac{f_{i_{m+1},i_n} + f_{i_{m+1},i_n-1} - f_{i_{m-1},i_n} - f_{i_{m-1},i_n-1}}{\Delta z^n_{i_n+1/2} + \Delta z^n_{i_n-1/2}}, \tag{2.3.16}$$

which, unlike Equation (2.3.15), does not preserve the monotonicity of f.

Advection interpolation

In the finite volume method, f is represented by its cell-average value on each cell of the distribution grid, so that the advective flux entering Equation (2.3.14) at a face requires a reconstruction of the face value $f_{i-1/2}$ from these cell averages,

$$f_{i-1/2} = \sum_{k=-2}^{1} \delta_k^{(i)} f_{i+k}, \tag{2.3.17}$$

for a set of reconstruction coefficients $\delta_k^{(i)}$. The simplest choice is the centered scheme, $\delta_{-1}^{(i)} = \delta_0^{(i)} = 1/2$, used by default, but is well known to produce spurious, unphysical oscillations for strongly advective problems. DREAM therefore also provides a range of positivity-preserving reconstruction schemes, from simple first- and second-order upwind schemes to non-linear flux-limiter schemes such as MUSCL [50], SMART [51] and TCDF [52], the last of which is generally recommended [16]. These flux limiters interpolate between a low-order (diffusive but monotone) and a high-order (more accurate but oscillatory) reconstruction, based on the local smoothness of the solution, so as to suppress spurious oscillations while retaining high-order accuracy where the solution is smooth. The non-linear flux limiter schemes require the use of the Newton (non-linear) solver described in Subsection 2.3.2.

Radial grid

The computational grid in DREAM is divided into $N_r \times N_p \times N_\xi$ cells, where the N_r points correspond to the radial grid representing the flux surfaces of the plasma, while N_p and N_ξ discretize the momentum-space representation of the electron distribution, as described below. The radial grid itself can be specified either analytically, or numerically, from an externally computed equilibrium. In this thesis, only numerical radial grids are used, obtained directly from the JOREK equilibrium, and only these are presented here.

The numerical grid is prescribed using the so-called LUKE format, adopted from the bounce-averaged Fokker-Planck solver LUKE [53], by specifying the following quantities:

- The major radius and vertical position of the magnetic axis, $(R_{\mathrm{ax}}, Z_{\mathrm{ax}})$.
- The poloidal flux grid $\boldsymbol{\Psi} = [\psi_1, \ldots, \psi_{n_\psi}]$, normalized by the plasma aspect ratio, $\psi_i \equiv \psi_{\mathrm{p},i}/(R_{\mathrm{ax}}/a)$, with a the plasma minor radius.
- The poloidal angle grid $\boldsymbol{\Theta} = [\theta_1, \ldots, \theta_{n_\theta}]$.
- The (R, Z) positions corresponding to each combination of $\boldsymbol{\Psi}$ and $\boldsymbol{\Theta}$: $\boldsymbol{X}, \boldsymbol{Y} \in \mathbb{R}^{n_\psi \times n_\theta}$, with $\boldsymbol{X}_{i,j} = R_{i,j} - R_0$ and $\boldsymbol{Y}_{i,j} = Z_{i,j} - Z_0$, where $R_{i,j}$ and $Z_{i,j}$ are the coordinates of the i-th flux surface at the j-th poloidal angle.
- The magnetic field components at the same grid points, $\boldsymbol{B_r}, \boldsymbol{B_z}, \boldsymbol{B_\varphi} \in \mathbb{R}^{n_\psi \times n_\theta}$, with $(\boldsymbol{B_r})_{i,j} = B_r(\psi_i, \theta_j)$, and analogously for $\boldsymbol{B_z}$ and $\boldsymbol{B_\varphi}$.

From this data, DREAM derives and tabulates all bounce and flux-surface averages and the Jacobian $\mathcal{V}'$ needed to evaluate Equations (2.3.1). A restriction of this method is that every magnetic field line must have exactly one minimum and one maximum in field strength along the flux surface; equilibria with more extrema (e.g. arising from very strong shaping) cannot be used in kinetic simulations [16]. An illustration of a numerical radial grid arising in a DREAM simulation is shown in Figure 2.8.

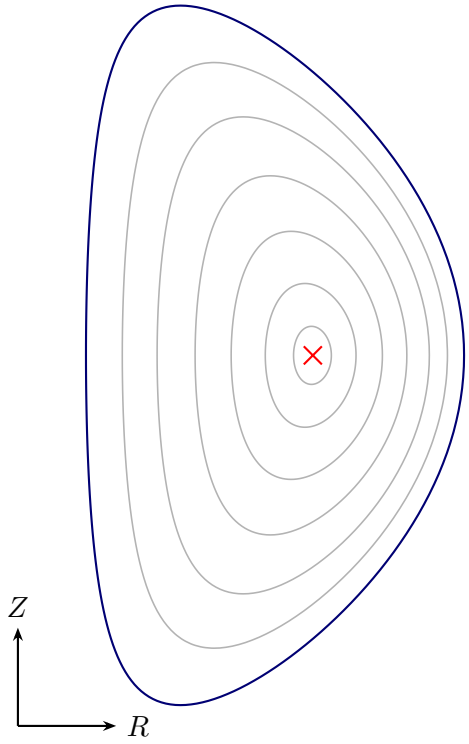


Figure 2.8: Illustration of the numerical radial grid in DREAM for an arbitrary equilibrium. The red cross denotes the magnetic axis. The blue surface represents the tokamak wall.

Momentum grid

As discussed in Subsection 2.3.1, the hot and runaway electron populations are resolved on two independent momentum grids in the fully kinetic model, since the hot and the

runaway population have different resolution requirements in p and ξ. As for the radial grid, each momentum grid consists of an interleaved flux and distribution grid.

In toroidal geometry, particles with pitch $|\xi_0| < \xi_T = \sqrt{1 - B_{\min}/B_{\max}}$ are trapped and bounce back and forth along the field line, while $|\xi_0| > \xi_T$ corresponds to passing particles. Since the bounce-averaged coefficients of Equation (2.3.1) are discontinuous (or singular) across $\xi_0 = \pm\xi_T$, accurately resolving f_{hot} and f_{re} in toroidal geometry requires grid points to be placed very close to the trapped-passing boundary at every radius. DREAM provides a dedicated grid type which automatically locates $\xi_T(r)$ during initialization and places a prescribed number of points in the trapped and passing regions, with a controllable boundary-layer width around ξ_T itself [16]. Without this refinement, the coarse resolution across the trapped-passing boundary can produce artificial oscillations in moments of the distribution function.

An illustration of the hot and runaway grids and their shared interface is shown in Figure 2.9.

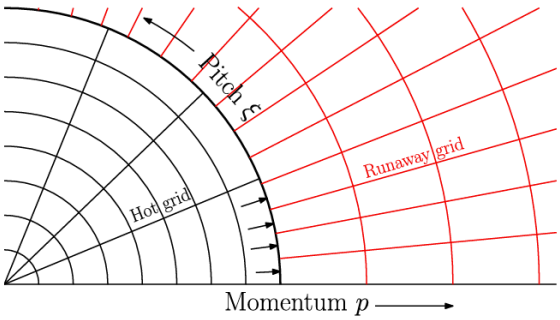


Figure 2.9: Kinetic momentum grids in DREAM: the hot grid (black) and the runaway grid (red), with the shared interface at $p = p_{re}$. Reprint from [16]

Boundary condition on the transition from the hot to the runaway region

Since f_{hot} and f_{re} are defined on separate, generally differently-resolved momentum grids, particles crossing $p = p_{\text{re}}$ must be transferred between them in a way that conserves particle number. This is enforced through a boundary condition based on the local density conservation equation

$$\Phi^{(p),\text{RE}}\left(p^{\text{RE}}_{1/2}, \xi^{\text{RE}}_J\right)\mathcal{V}'^{\text{RE}}_{1/2,J}\Delta\xi^{\text{RE}}_J = -\sum_j \Phi^{(p),\text{hot}}\left(p^{\text{hot}}_{\max}, \xi^{\text{hot}}_j\right)\mathcal{V}'^{\text{hot}}_{N_p+1/2,j}\overline{\Delta\xi}_{jJ}, \tag{2.3.18}$$

where $\Phi^{(p),\text{RE}}$ and $\Phi^{(p),\text{hot}}$ are the particle fluxes into the runaway grid and out of the hot grid respectively, and the overlap between cells j (hot grid) and J (runaway grid) in pitch is

$$\overline{\Delta\xi}_{jJ} = \min\left(\xi^{\text{hot}}_{j+1/2}, \xi^{\text{RE}}_{J+1/2}\right) - \max\left(\xi^{\text{hot}}_{j-1/2}, \xi^{\text{RE}}_{J-1/2}\right). \tag{2.3.19}$$

Requiring the flux of particles to be locally conserved as in Equation (2.3.18) yields, for the advective and diffusive fluxes on the hot electron grid,

$$\Phi^{(p),\text{hot}}_{\text{adv}}\left(p^{\text{hot}}_{N_p+1/2}, \xi^{\text{hot}}_j\right) = F^{\text{hot}}_{N_p+1/2,j} f^{\text{hot}}_{N_p+1/2,j} = F^{\text{hot}}_{N_p+1/2,j}\left[\delta^{(1)}_j f^{\text{hot}}_{N_p,j} + (1-\delta^{(1)}_j)\hat{f}^{\text{RE}}_{1,j}\right], \tag{2.3.20}$$

$$\Phi^{(p),\text{hot}}_{\text{diff}}\left(p^{\text{hot}}_{N_p+1/2}, \xi^{\text{hot}}_j\right) = D^{\text{hot}}_{N_p+1/2,j}\left.\frac{\partial f^{\text{hot}}}{\partial p}\right|_{N_p+1/2,j} = D^{\text{hot}}_{N_p+1/2,j}\frac{f^{\text{hot}}_j - \hat{f}^{\text{RE}}_{1,j}}{p^{\text{RE}}_{1/2} - p^{\text{hot}}_{N_p+1/2}}, \tag{2.3.21}$$

with the averaged runaway distribution function

$$\hat{f}_{1,j}^{\mathrm{RE}} = f_{J_j}^{\mathrm{RE}} + \left(\xi_j^{\mathrm{hot}} - \xi_{J_j-1}^{\mathrm{RE}}\right) \frac{f_{J_j}^{\mathrm{RE}} - f_{J_j-1}^{\mathrm{RE}}}{\xi_{J_j}^{\mathrm{RE}} - \xi_{J_j-1}^{\mathrm{RE}}}, \tag{2.3.22}$$

where J_j denotes the smallest integer such that $\xi_{J_j}^{\mathrm{RE}} \geq \xi_j^{\mathrm{hot}}$, and the interpolation coefficients $\delta_j^{(1)}$ are chosen with an upwind scheme,

$$\delta_j^{(1)} = \begin{cases} 1, & \text{if } F_{N_p+1/2,j}^{\mathrm{hot}} \leq 0, \\ 0, & \text{otherwise,} \end{cases} \tag{2.3.23}$$

to minimize spurious oscillations at the grid boundary. The fluxes on the runaway grid then follow by combining Equations (2.3.18)-(2.3.21),

$$\Phi_{1/2,J}^{(p),\mathrm{RE}} = \frac{-\sum_j \Phi_{N_p+1/2,j}^{(p),\mathrm{hot}} \mathcal{V}_{N_p+1/2,j}^{\mathrm{hot}} \overline{\Delta\xi}_{jJ}}{\mathcal{V}_{1/2,J}^{\mathrm{RE}} \Delta\xi_J^{\mathrm{RE}}}. \tag{2.3.24}$$

Time evolution

As DREAM uses an implicit Euler time discretization, at every time step a generally non-linear system of equations must be solved. By moving all terms in Equation (2.3.14) to the same side, a system of equations is obtained, for time $t = t_{l+1}$, which is denoted by the operator $\boldsymbol{F}$:

$$\boldsymbol{F}\left(\boldsymbol{u}^{(l+1)}\right) = \boldsymbol{0} \tag{2.3.25}$$

for the vector of unknowns $\boldsymbol{u}$. Two solvers are available for this purpose.

The non-linear (Newton) solver linearizes $\boldsymbol{F}$ around successive estimates $\boldsymbol{u}_i^{(l+1)}$ of the solution, giving the iterative scheme

$$\boldsymbol{u}_{i+1}^{(l+1)} = \boldsymbol{u}_i^{(l+1)} - \mathsf{J}^{-1}\left(\boldsymbol{u}_i^{(l+1)}\right) \boldsymbol{F}\left(\boldsymbol{u}_i^{(l+1)}\right), \tag{2.3.26}$$

where the index i denotes the Newton iteration and J is the Jacobian of $\boldsymbol{F}$. In practice, the Newton step $\Delta\boldsymbol{u}_i^{(l+1)} = \boldsymbol{u}_{i+1}^{(l+1)} - \boldsymbol{u}_i^{(l+1)}$ is obtained by solving the sparse linear system $\mathsf{J}\left(\boldsymbol{u}_i^{(l+1)}\right)\Delta\boldsymbol{u}_i^{(l+1)} = \boldsymbol{F}\left(\boldsymbol{u}_i^{(l+1)}\right)$ rather than by explicitly forming J^{-1}, and this system is re-assembled and re-solved at every iteration i, until the change in each unknown quantity falls below a prescribed absolute and relative tolerance. Rather than the exact derivative of $\boldsymbol{F}$, an approximate Jacobian is used: some derivatives are simplified or neglected altogether, both for computational efficiency and to preserve the sparsity of the resulting matrix. This solver is generally the more robust of the two, particularly for rapidly time-varying or strongly non-linear scenarios such as a thermal quench.

The linearly implicit solver instead linearizes the full equation system in time around the current time step,

$$\boldsymbol{F}\left(\boldsymbol{u}^{(l+1)}\right) \approx \mathsf{M}\left(\boldsymbol{u}^{(l)}\right)\boldsymbol{u}^{(l+1)} + \boldsymbol{S}\left(\boldsymbol{u}^{(l)}\right), \tag{2.3.27}$$

where $\mathsf{M}\left(\boldsymbol{u}^{(l)}\right)$ is a matrix representing the discretized operators found in the discretization (2.3.14) and $\boldsymbol{S}\left(\boldsymbol{u}^{(l)}\right)$ is a vector representing the sources S in the equation. With this method, the solution at the next time step is obtained by solving the single sparse linear system $\mathsf{M}\left(\boldsymbol{u}^{(l)}\right)\boldsymbol{u}^{(l+1)} = -\boldsymbol{S}\left(\boldsymbol{u}^{(l)}\right)$, without any iteration. This makes the linearly

implicit solver substantially faster per time step, and is suitable for linear and nonlinear systems which vary slowly in time.

Both solvers rely on third-party sparse linear solvers from the PETSc library [54, 55]: the Newton solver requires one sparse linear solve per iteration, while the linearly implicit solver requires a single sparse linear solve per time step.

Chapter 3

Coupling of JOREK and DREAM

Chapter 1 introduced the coupling of JOREK and DREAM as the central aim of this thesis project, to build a new RE modeling framework that simultaneously can capture the free boundary plasma dynamics and the kinetic RE phase space dynamics. No such coupling existed prior to this work: the framework, numerical methods, and implementation presented in this chapter were developed entirely within the scope of this thesis, rather than building on or extending any pre-existing JOREK-DREAM interface. Section 3.1 begins by setting the JOREK-DREAM coupling into the broader research picture. Next, Section 3.2 gives an overview of the workflow of the JOREK-DREAM coupling. Finally, Section 3.3 details the newly developed and implemented numerical algorithms used for coupling the two codes.

3.1 Motivation for coupling JOREK and DREAM

Having covered both the physics of the major disruption phases and the physical and numerical models of each code, the coupling will now be detailed and its fit into the broader modeling framework will be explained.

JOREK's nonlinear finite-element treatment of the MHD equations (Section 2.2.2) lets it evolve the plasma in a free boundary setting through its coupling to STARWALL [35] or CARIDDI [36], taking into account the time-evolution of currents in external conductors and their self-consistent two-way interaction with the plasma dynamics. Via its kinetic particle module, JOREK can also evolve the full momentum-space distribution of the runaway electrons themselves. This high-fidelity approach comes at a steep computational cost. Even with the substantial speedup demonstrated by porting the 3D kinetic module to GPUs [56], simulating the full disruption event while tracking the RE distribution function using the Monte Carlo (PIC) kinetic approach in JOREK remains computationally intractable for the long phases of RE generation, vertical plasma motion, and RE plateau phase, which for ITER-scale disruptions is expected to reach the multi-second range – several orders of magnitude longer than the millisecond-scale TQ that precedes them or the RE termination that typically unfolds on the order of tens of microseconds when REs are finally lost in a burst of MHD activity. As such, the JOREK 3D kinetic module is particularly suitable for short phases with strong MHD activity, like the TQ (including hot-tail seed formation [57]) or the termination.

DREAM takes the opposite trade-off. Its 1D flux surface averaged description significantly lowers the dimensionality of the problem by not considering the self-consistent evolution of the 2D poloidal cross-section with a free evolving plasma boundary and realistic wall currents or any 3D MHD dynamics. RE transport is incorporated via externally

provided advection-diffusion coefficients. These modeling assumptions allow DREAM to evolve the phase-space distribution (p, ξ) of REs across the long time scales of RE generation during the CQ and plateau phases. Since the plasma equilibrium is not evolved, DREAM uses a fixed flux surface geometry that is given as an input by the user, e.g., based on experimental equilibrium reconstructions [58]. Therefore, DREAM is particularly suitable for RE formation studies in vertically stable plasmas. For vertically unstable plasmas or MHD active CQ dynamics, it requires externally provided information.

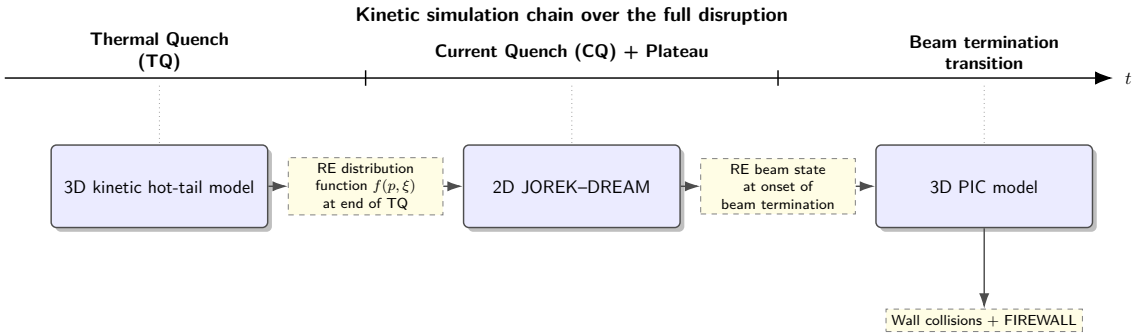


Figure 3.1: Envisioned simulation workflow capturing the kinetic evolution of runaway electrons throughout the full disruption event. The aim of the JOREK-DREAM coupling is to handle the kinetic evolution over the very long CQ and plateau time scales.

There is a gap in the model hierarchy that this thesis aims to close by coupling the two. In the chosen approach, JOREK evolves the background plasma and the free-boundary equilibrium in space and time, supplying the input DREAM needs to evolve the runaway electron distribution function kinetically in one dimension in space and two dimensions in phase space. DREAM then supplies JOREK with the runaway current density, allowing JOREK to accurately compute the evolution of the electric field and the entire plasma dynamics. The two codes exchange this information at regular intervals during the simulation (for the work presented in this thesis, this occurs at every JOREK fluid time step). This establishes a new model that does not address 3D MHD dynamics, which require standalone JOREK simulations, but treats everything else self-consistently, in particular the interplay between current decay, vertical plasma motion, and RE formation and plateauing. The model thus reaches the limits of validity in situations where the CQ is strongly MHD active, but it can treat an MHD-quiet CQ fully self-consistently, providing information about the RE plateau state that other physics models cannot provide and experiments struggle to measure. This enables entirely new physics studies related to disruption physics and disruption mitigation by itself, but can also be readily combined with the different fidelity models that JOREK and DREAM offer themselves, by switching to another fidelity level during the dynamics as needed.

An example of such a transition lies in the RE formation. The post-TQ RE population is expected to be dominated by the hot-tail mechanism in an ITER-like plasma (at least in the non-nuclear phase), where the effect of 3D magnetic field perturbations on particle transport can play an essential role. A model to accurately capture the hot-tail mechanism kinetically during the TQ phase in JOREK was recently developed [57]. This high-fidelity approach can provide very accurate information about the RE population after the TQ. Such a state can be taken as starting point for a JOREK-DREAM coupled simulation of the CQ. The following chain could be applied in future studies: the 3D JOREK model provides the hot-tail seed, JOREK-DREAM then takes over for the CQ, vertical motion and plateau, and when the assumption of an MHD-quiet background plasma breaks down

close to the beam termination, the study can be continued in JOREK with the 3D PIC model coupled to the 3D plasma dynamics. In addition, a kinetic model for the avalanche source term was recently implemented into the JOREK 3D kinetic model [59] (again computationally far too expensive for CQ studies), allowing to address MHD-active re-avalanching phases for an RE beam after termination. JOREK also includes a wall collision module [28, 29], allowing to assess the impact of RE kinetic markers onto realistic 3D CAD models of the tokamak first wall, and the volumetric wall heating can be efficiently computed using the newly developed FIREWALL code [60]. The full modeling chain is visualized in Figure 3.1. This thesis contributes to it by establishing the missing gap between TQ and termination, where so far a choice had to be made whether to capture the self-consistent plasma evolution and vertical motion in realistic geometries (using the JOREK RE fluid model) or retaining the knowledge about the RE phase-space distribution (using DREAM).

3.2 Workflow of the coupling

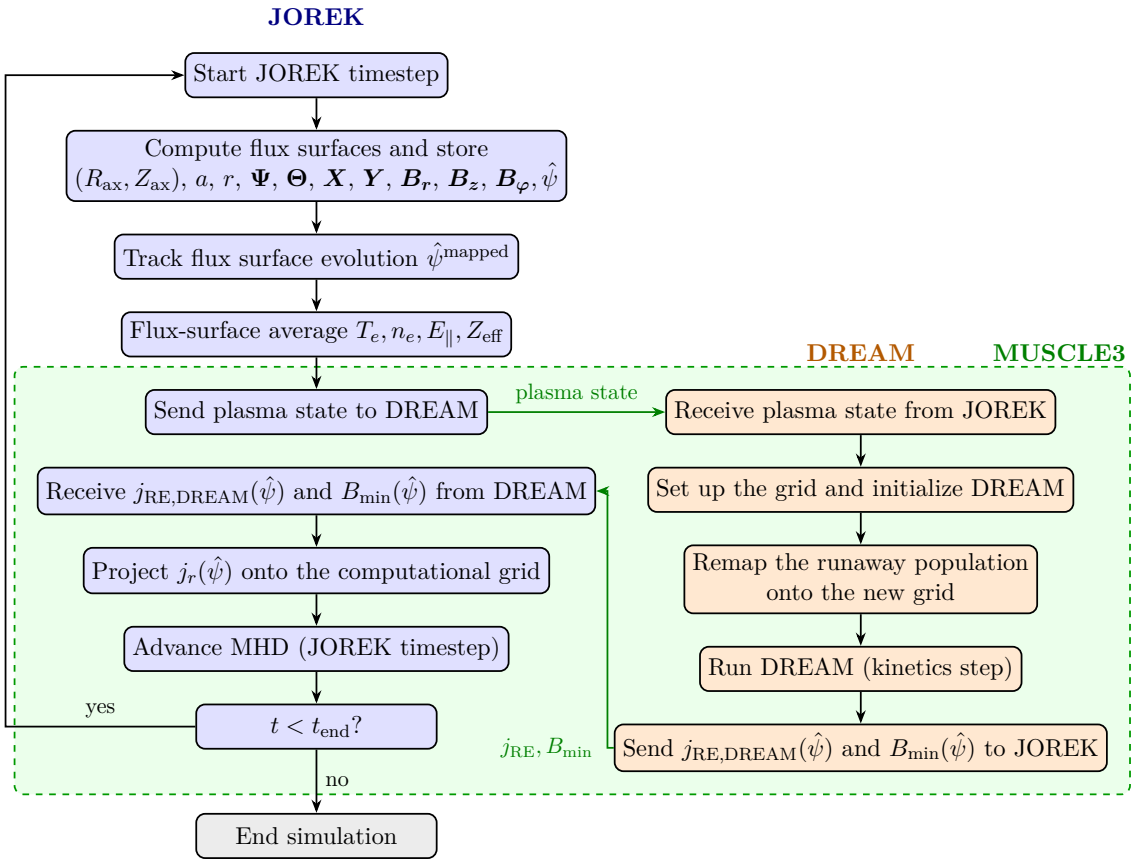


Figure 3.2: Illustration of the workflow of the coupling between JOREK and DREAM using MUSCLE3. The green dashed box represents the part of the simulation handled by MUSCLE3.

In this thesis, the two-way coupling is executed at each JOREK fluid time step; if deemed beneficial, the interval can be extended so that communication occurs only after a set number of fluid time steps. This section gives an overview of the general coupling procedure, illustrated in Figure 3.2, while the numerical implementation is detailed in Sec-

tion 3.3. The two codes run as separate programs and exchange data through MUSCLE3, a framework for coupling simulation codes in multiscale simulations [61], whose use in this work is described in Section 3.3.1. The left column of the figure corresponds to JOREK, the right one to DREAM, and the green box marks the part of the process handled by MUSCLE3. One coupling cycle proceeds as follows.

1. **JOREK extracts and averages the plasma state.** After configuring the initial JOREK equilibrium and computational grid, the coupling requires the user to specify the number of flux surfaces n_ψ on which JOREK evaluates the equilibrium and background profiles at each coupling step. At the start of each JOREK fluid step, the magnetic geometry is extracted on n_ψ flux surfaces, chosen large enough to accurately represent the geometry, and the numerical radial grid is constructed in the LUKE format required by DREAM (Section 2.3.2). The same n_ψ flux surfaces are then used to compute the flux-surface averages of the electron temperature T_e, the free electron density n_e, the parallel electric field $E_\parallel$, and the effective charge Z_{eff}. Both the equilibrium geometry and these averaged profiles are evaluated on the same fixed, uniform grid of normalized poloidal flux values $\{\hat{\psi}_k\}_{k=1}^{n_\psi}$, unchanged from step to step, so that DREAM's domain always covers the full closed-flux-surface region and receives geometry and profiles defined on a mutually consistent grid. This grid is unrelated to DREAM's own internal radial solve grid, with N_r cells, onto which DREAM independently interpolates the received geometry and profiles (Section 2.3.2); n_ψ can therefore be chosen much larger than N_r, to represent the equilibrium and profiles continuously, without affecting the resolution or cost of DREAM's own solve. Since JOREK's fixed $\{\hat{\psi}_k\}$ grid does not itself move with the plasma, a separate mapping $\hat{\psi}^{\text{mapped}}$ is also computed and sent, so that DREAM can conservatively remap its own state, on its own grid, onto the new step's fixed flux labels (Section 3.3).

2. **JOREK sends the plasma state to DREAM.** The full set of quantities sent at each coupling step is listed in Table 3.1. JOREK then remains idle while DREAM advances.

3. **DREAM evolves the runaway electron population.** DREAM builds its magnetic equilibrium from the received LUKE data, initializes the simulation using the prescribed profiles, remaps the electron population from its previous run onto the new grid, and evolves it for the duration of the JOREK fluid time step. The resulting electron distribution function is retained for the next DREAM run, while the radial profile of the runaway current density $j_{\text{RE,DREAM}}$ and the minimum magnetic field $B_{\min}$ on each flux surface are sent back to JOREK, as summarized in Table 3.2.

4. **JOREK incorporates the runaway current and advances.** JOREK projects j_r onto its computational grid and includes it in the induction equation (2.2.9) (Section 3.3), then advances the next fluid time step.

This cycle repeats until the simulation reaches the desired end time.

3.3 Numerical implementation of the coupling

The numerical implementation of the coupling will now be detailed. To start, the MUSCLE3 framework and its implementation for the coupling is briefly explained in Section 3.3.1. Then, in Section 3.3.2, the algorithms implemented to compute the magnetic

Table 3.1: Simulation data sent from JOREK to DREAM at each coupling step. Quantities of dimension n_ψ are radial profiles on the flux-surface grid.

Field	Symbol	Dimensions
Step index	`istep`	scalar
Coupling time step	t_{fluid}	scalar
Electron temperature	T_e	n_ψ
Electron density	n_e	n_ψ
Parallel electric field	$E_\parallel$	n_ψ
Effective charge	Z_{eff}	n_ψ
Magnetic-axis position	$(R_{\text{ax}}, Z_{\text{ax}})$	scalar×2
Minor radius	a	scalar
Midplane radial coordinate	r	n_ψ
Poloidal flux (absolute)	Ψ	n_ψ
Poloidal angle array	$\boldsymbol{\Theta}$	n_θ
R, Z on flux-surface grid	$\boldsymbol{X}, \boldsymbol{Y}$	$n_\psi \times n_\theta$
Magnetic field components	$\boldsymbol{B_r}, \boldsymbol{B_z}, \boldsymbol{B_\varphi}$	$n_\psi \times n_\theta$
Normalized poloidal flux (fixed grid)	$\hat{\psi}$	n_ψ
Mapped normalized poloidal flux	$\hat{\psi}^{\text{mapped}}$	n_ψ

Table 3.2: Simulation data sent from DREAM to JOREK at each coupling step. Both quantities are radial profiles on the flux-surface grid.

Field	Symbol	Dimensions
Runaway electron current density	$j_{\text{RE,DREAM}}$	n_ψ
Minimum of the magnetic field	$B_{\min}$	n_ψ

equilibrium and its evolution in JOREK in between fluid steps is described. Next, the remapping of the electron distribution from one DREAM run to the next is addressed in Section 3.3.3. Finally, the projection of the runaway current density onto JOREK's computational grid is described in Section 3.3.4. These sections are presented in the order in which their respective steps occur within a single coupling cycle.

3.3.1 The multiscale coupling framework MUSCLE3

JOREK and DREAM are coupled using MUSCLE3 [62, 63, 61], an open-source framework for multiscale simulations developed by the Netherlands eScience Center and the University of Amsterdam. A multiscale simulation couples several submodels that operate at different spatial or temporal scales and exchange information during the run. MUSCLE3 provides the software infrastructure for this. In MUSCLE3 each submodel (a component, in MUSCLE3 terminology) communicates through ports, sending and receiving messages, where the coupling topology is defined externally in a YAML-based configuration file called a `yMMSL` file. The components are linked against the MUSCLE3 library `libmuscle`, which provides the `send()`/`receive()` primitives and is available for Python, C++ and Fortran; this is what allows JOREK (Fortran) and DREAM (C++/Python) to be coupled without extensive modification of either code. The different components are coordinated by the central run-time component `muscle_manager`, which the components use to find each other in the communication.

The pattern used in this thesis is MUSCLE3's macro-micro coupling framework [62]: a coarse-grained (macro) submodel is advanced one time step at a time, and within each step a fine-grained (micro) submodel is used to evolve another quantity that is too expensive,

or impossible, to compute within the macro model itself. JOREK acts as the macro submodel, advancing the MHD plasma state, while DREAM acts as the micro submodel, evolving the runaway electron distribution and returning the runaway current density $j_{\mathrm{RE,DREAM}}$ used to handle the resistive decay of the electric field in the induction equation (**??**).

The resulting coupling configuration is given in the `yMMSL` file of Listing 3.1. It defines two components, `jorek` and `dream`, connected by two conduits, one for each direction of the data exchange. Each port is attached to an operator, which specifies at which point of the component's time loop the communication takes place, and it is this choice of operators that realises the macro-micro pattern. On the JOREK side, the plasma state is sent through `plasma_state_out` under the intermediate-output operator `o_i`, i.e. during each fluid time step, and the runaway current is received through `jre_in` under the state-update operator `s`, before the fluid state is advanced. On the DREAM side, the plasma state is received through `plasma_state_in` under `f_init`, which initializes each DREAM run, and the result is returned through `jre_out` under the final-output operator `o_f` once the run has completed. As a result, DREAM performs one complete run for every JOREK fluid time step, as illustrated in Figure 3.2.

Listing 3.1: `yMMSL` configuration for the JOREK-DREAM coupling.

```
ymmsl_version: v0.1
model:
  name: jorek_dream_coupling
  components:
    jorek:
      implementation: jorek
      ports:
        o_i: [plasma_state_out]
        s:   [jre_in]
    dream:
      implementation: dream
      ports:
        f_init: [plasma_state_in]
        o_f:    [jre_out]
  conduits:
    jorek.plasma_state_out: dream.plasma_state_in
    dream.jre_out:          jorek.jre_in
```

3.3.2 Tracking the evolution of flux surfaces in JOREK

During a disruption event, the flux surfaces are always evolving and without further measures this could introduce an unphysical RE transport in the coupled model. JOREK evolves the magnetic equilibrium self-consistently through the poloidal flux ψ, governed by the induction equation (2.2.9) as part of the reduced MHD system (2.2.9)–(2.2.14) introduced in Section 2.2. However, as mentioned earlier, DREAM assumes a fixed equilibrium in a given simulation. For this reason, at every coupling step, JOREK hands off simulation data to DREAM, where the magnetic equilibrium is reconstructed anew using the LUKE file format as described in Section 2.3. Furthermore, the runaway population in DREAM needs to be remapped from one DREAM run to the next, as the magnetic equilibrium changes from one run to the other. This requires tracking the RE population's position with respect to the flux surfaces within one JOREK step, i.e. keeping track of the radial coordinate changes throughout a JOREK time step. To begin with, the algorithm

used to compute the flux surface geometry in JOREK will be detailed, after which the methodology used to track their evolution will be explained.

Computing the flux surface geometry

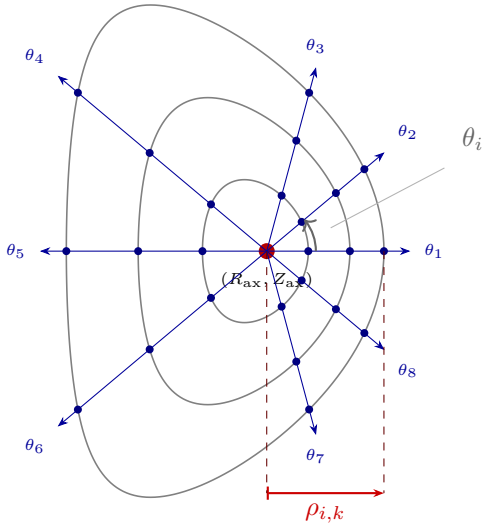


Figure 3.3: Illustration of the ray-tracing root-finding algorithm 1.

In the DREAM numerical radial grid, flux surfaces are described by their (R, Z) coordinates and by their poloidal flux ψ values and poloidal angle values θ (Section 2.3.2). These are computed in JOREK using a ray-tracing root-finding approach. This method avoids the computational complexity of geometric contour-tracing by solving directly for surface coordinates on a predefined angular grid, exploiting the fact that for a tokamak plasma the plasma boundary is a convex surface.

A uniform grid of n_θ poloidal angles is defined as

$$\theta_i = \frac{i}{n_\theta} \cdot 2\pi, \qquad i = 0, \ldots, n_\theta, \tag{3.3.1}$$

where the interval is $[0, 2\pi]$ so that the first and last points are coincident. For each angle θ_i, a ray from the magnetic axis $(R_{\mathrm{ax}}, Z_{\mathrm{ax}})$ is parametrised by the geometric distance ρ from the axis:

$$R(\rho, \theta_i) = R_{\mathrm{ax}} + \rho \cos\theta_i, \tag{3.3.2}$$
$$Z(\rho, \theta_i) = Z_{\mathrm{ax}} + \rho \sin\theta_i. \tag{3.3.3}$$

The position of the magnetic axis $(R_{\mathrm{ax}}, Z_{\mathrm{ax}})$ is already derived in JOREK and it can be assumed that it is given. For a target flux surface k with normalised poloidal flux $\hat{\psi}_k \in (0, 1)$, the target poloidal flux value is

$$\psi_k^{\mathrm{target}} = \psi_{\mathrm{axis}} + \hat{\psi}_k \left(\psi_{\mathrm{bnd}} - \psi_{\mathrm{axis}}\right). \tag{3.3.4}$$

Along ray i at fixed θ_i, the coordinate $\rho_{i,k}$ of surface k is found as the root of the one-dimensional residual

$$f(\rho) = \psi(R(\rho, \theta_i),\, Z(\rho, \theta_i)) - \psi_k^{\mathrm{target}}, \tag{3.3.5}$$

using the secant method, which approximates the local derivative using two successive iterates $\rho^{(n-1)}$ and $\rho^{(n)}$:

$$\rho^{(n+1)} = \rho^{(n)} - f\left(\rho^{(n)}\right) \frac{\rho^{(n)} - \rho^{(n-1)}}{f\left(\rho^{(n)}\right) - f\left(\rho^{(n-1)}\right)}. \tag{3.3.6}$$

The full procedure, including the choice of initial guesses, the clamping of each iterate to a physically admissible interval, and the convergence criterion, is summarised in Algorithm 1 and is visualized in Figure 3.3.

Algorithm 1 Root-finding for a single flux-surface coordinate $(R_{i,k}, Z_{i,k})$

Input: angle θ_i, target flux ψ_k^{target}, magnetic axis $(R_{\text{ax}}, Z_{\text{ax}})$, minor radius of the last closed flux surface a_{LCFS}, warm-start radius ρ_{prev} from the previous surface $k-1$ at the same angle (if $k > 1$).
if $k = 1$ **then**
    $\rho^{(0)} \leftarrow 0, \quad \rho^{(1)} \leftarrow 0.02$ m
else if $\rho_{\text{prev}} \geq 10^{-2}$ m **then**
    $\rho^{(0)} \leftarrow 0.98\,\rho_{\text{prev}}, \quad \rho^{(1)} \leftarrow \rho_{\text{prev}}$
else
    $\rho^{(0)} \leftarrow 0.8\sqrt{\hat{\psi}_k}\,a_{\text{LCFS}}, \quad \rho^{(1)} \leftarrow \sqrt{\hat{\psi}_k}\,a_{\text{LCFS}}$
end if
repeat
    Update $\rho^{(n+1)}$ via the secant step, Equation (3.3.6).
    Clamp: $\rho^{(n+1)} \leftarrow \max\left(0,\ \min\left(\rho^{(n+1)},\ 2\sqrt{\hat{\psi}_k}\,a_{\text{LCFS}}\right)\right)$
until $\left|\rho^{(n+1)} - \rho^{(n)}\right| < 10^{-8}$ m
Output: $(R_{i,k}, Z_{i,k}) = \left(R_{\text{ax}} + \rho_{i,k}\cos\theta_i,\ Z_{\text{ax}} + \rho_{i,k}\sin\theta_i\right)$.

Applying Algorithm 1 at every angle θ_i and flux-surface index k builds the output matrices $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^{n_\theta \times n_\psi}$ needed to construct the numerical radial grid in DREAM (Section 2.3.2).

After the flux surface coordinates (R, Z) have been computed using this ray-tracing root-finding approach, it is now possible to compute the magnetic field components B_r, B_z and B_φ at these points, using existing JOREK functionalities [14], building the matrices $\boldsymbol{B_r}$, $\boldsymbol{B_z}$ and $\boldsymbol{B_\varphi}$.

Flux-surface mapping in between coupling steps

To track the evolution of the flux surfaces within the JOREK time step, GC markers with low kinetic energy and pitch angle are tracked over its duration, as particles with these properties follow magnetic field lines closely including also their vertical motion, so tracking them provides a direct way to observe how the field lines, and consequently the flux surfaces evolve.

One GC marker is initialized on each flux surface. Each marker is placed at the point where its surface crosses the outboard midplane, i.e. at $\theta = 0$. Concretely, the marker associated with flux surface s is placed at

$$\mathbf{x}_s^n = \left(R_{\text{ax}} + r_s^n(t_n),\ Z_{\text{ax}},\ 0\right),$$

where $r_s^n(t_n)$ denotes the minor-radius distance from the magnetic axis to the point where the flux surface s intersects the outboard midplane at time t_n. Further, the markers are

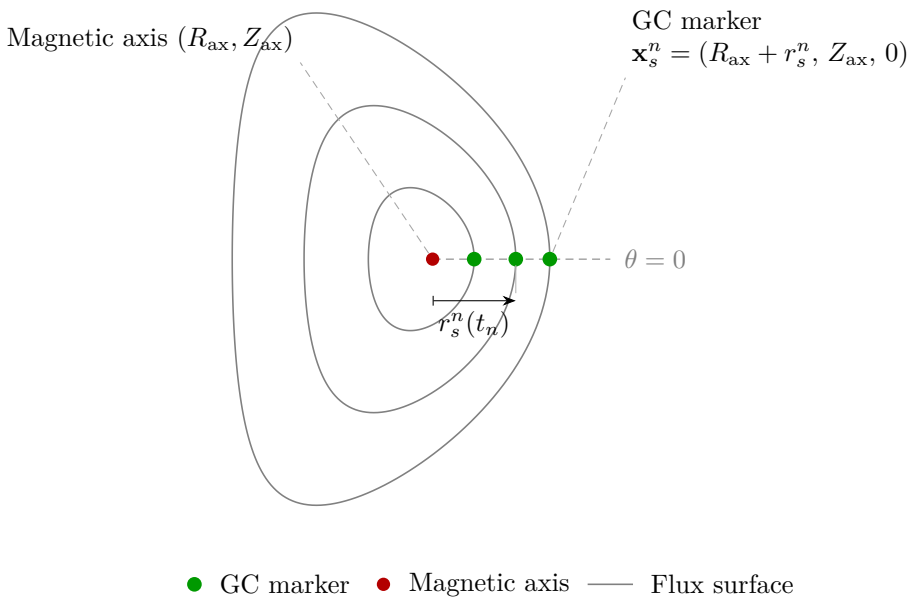


Figure 3.4: Initialization of guiding-center (GC) markers. One marker is placed on each flux surface at its intersection with the outboard midplane ($\theta = 0$), i.e. at a distance $r_s^n(t_n)$ from the magnetic axis along the midplane.

initialized with an energy of around 1 keV (low energy to ensure they follow the flux surfaces well) and a pitch of $\xi = \frac{p}{p_\parallel} = 0.999$ (almost exactly parallel to the magnetic field line). The exact properties of these markers could be changed in the future depending on the needs. The initialization of the particles is illustrated in Figure 3.4.

Each marker is then pushed forward in JOREK's own electromagnetic fields throughout the fluid time step using a relativistic guiding-centre integrator (fifth-order Cash-Karp Runge-Kutta) as explained in Section 2.2.1. The fields are interpolated in time between the equilibrium at the start and end of the step so that the marker experiences the field's actual evolution rather than a frozen snapshot. At the marker's final position, the normalized poloidal flux is evaluated, giving the mapped value $\hat{\psi}_k^{\text{mapped}}$, which corresponds to the normalized flux value that the plasma starting this step at $\hat{\psi}_k$ has moved to by the end of it. The procedure is summarised in Algorithm 2.

Algorithm 2 flux-surface mapping

1: **Input:** Fluid time step $[t_n, t_{n+1}]$, fixed normalised flux grid $[\hat{\psi}_1, \ldots, \hat{\psi}_{n_\psi}]$ (identical every step).
2: **Initialisation:** Locate the flux-surface geometry of $\{\hat{\psi}_k\}$ in the equilibrium at t_n, and place n_ψ markers on the outboard midplane, at coordinates $\mathbf{x}_s^n = (R_{\mathrm{ax}} + r_s^n(t_n),\ Z_{\mathrm{ax}},\ 0)$, one per flux surface.
3: **for** each surface $s = 1, \ldots, n_\psi$ **do**
4: Advance marker s from t_n to t_{n+1}, obtaining end position $\mathbf{x}_s^{n+1} = (R_{\mathrm{end}}, Z_{\mathrm{end}}, \phi_{\mathrm{end}})$.
5: Evaluate the normalized poloidal flux at the end position, in the equilibrium at t_{n+1}: $\hat{\psi}_s^{\text{mapped}} = \hat{\psi}(\mathbf{x}_s^{n+1},\ t_{n+1})$.
6: **end for**
7: **Output:** Mapping $[\hat{\psi}_1^{\text{mapped}}, \ldots, \hat{\psi}_{n_\psi}^{\text{mapped}}]$, sent to DREAM alongside the (unchanged) fixed grid $\{\hat{\psi}_k\}$.

3.3.3 Remapping the runaway population in DREAM

Algorithm 3 Conservative remap of f_{hot} and f_{re}

1: **Input:** previous-step distribution $f^{\,n}(r,p,\xi)$ on the old DREAM grid, with radial cell volumes $V'^{\,n}\Delta r^{\,n}$; the mapping $\hat{\psi} \mapsto \hat{\psi}^{\mathrm{mapped}}$; the new DREAM grid's cell faces $r_f^{\,n+1}$ and volumes $V'^{\,n+1}\Delta r^{\,n+1}$.
2: Evaluate the flux label of each old-grid cell face in the old equilibrium, $\hat{\psi}(r_f^{\,n})$.
3: Push these labels through the tracer mapping to obtain $\hat{\psi}^{\mathrm{mapped}}(r_f^{\,n})$, i.e. where the material bounding each old cell actually ended up.
4: Evaluate the flux label of each new-grid cell face in the new equilibrium, $\hat{\psi}(r_f^{\,n+1})$.
5: **for each** momentum-grid point (p,ξ) **do**
6: Form the cumulative particle count enclosed by each old cell face, $N(\hat{\psi};p,\xi) = \sum_{k\le\cdot} f_k^{\,n}(p,\xi)\,V_k'^{\,n}\Delta r_k^{\,n}$, and interpolate it as a function of $\hat{\psi}^{\mathrm{mapped}}(r_f^{\,n})$ onto the new cell faces $\hat{\psi}(r_f^{\,n+1})$.
7: Set $f^{\,n+1}(r,p,\xi) = \Delta N(\cdot\,;p,\xi)/(V'^{\,n+1}\Delta r^{\,n+1})$, obtained by differentiating the interpolated cumulative count and dividing by the new cell volumes.
8: **end for**
9: **Output:** $f^{\,n+1}(r,p,\xi)$ on the new DREAM grid.

Since the equilibrium changes between coupling steps (ψ evolves through the induction equation (2.2.9)), a flux surface labelled $\hat{\psi}_k$ at step n does not, in general, coincide with the same physical surface at step $n+1$. The mapping $\hat{\psi}_k \mapsto \hat{\psi}_k^{\mathrm{mapped}}$ constructed in Section 3.3.2 quantifies this displacement. In DREAM, carrying f_{hot} and f_{re} over unchanged, cell for cell, would therefore introduce spurious transport or particle creation/destruction: the volume $\mathcal{V}_k'\Delta r_k$ of radial cell k, appearing in the finite-volume discretization of the drift-kinetic equation, Equation (2.3.14), changes as the equilibrium evolves, and the particles it contained need not have remained at that label. Instead, f_{hot} and f_{re} are advanced by remapping the particle count $f\,\mathcal{V}'\Delta r$ rather than the distribution function itself, which automatically accounts for such volumetric changes. The remap is carried out entirely on DREAM's radial grid, with JOREK's flux labels interpolated onto it, and the momentum coordinates (p,ξ) are unaffected by the new equilibrium and carried through unchanged. The remapping procedure is described in Algorithm 3 and is applied at every momentum-grid point. This algorithm is quasi-conservative, as it can lead to a small (but quantifiable) loss or gain of particles.

Analytic conservation property

Since the particle number should be (almost) conserved by Algorithm 3, it is necessary to establish that this is actually the case. This is done by proving Lemma 1.

Lemma 1. *Fix an arbitrary momentum-grid point (p,ξ), and let $f_k^{\,n}$ denote $f^{\,n}(r_k,p,\xi)$ at that point. Let the old grid (timestep n) have K cells with faces $r_{f,k}^{\,n}$, $k=0,\dots,K$, and volumes $V_k'^{\,n}\Delta r_k^{\,n}$, and let the new grid (timestep $n+1$), possibly with a different number of cells, have M cells with faces $r_{f,i}^{\,n+1}$, $i=0,\dots,M$, and volumes $V_i'^{\,n+1}\Delta r_i^{\,n+1}$. If $f^{\,n+1}$ is obtained from $f^{\,n}$ by Algorithm 3, then*

$$\sum_{i=0}^{M-1} f_i^{\,n+1}\,V_i'^{\,n+1}\Delta r_i^{\,n+1} \;=\; \sum_{k=0}^{K-1} f_k^{\,n}\,V_k'^{\,n}\Delta r_k^{\,n} \;-\; \Delta N_{\mathrm{lost}}, \tag{3.3.7}$$

where

$$\Delta N_{\mathrm{lost}} = N^{\,n}(K) - \mathcal{N}^{\,n}(\hat{\psi}(r_{f,M}^{\,n+1})), \tag{3.3.8}$$

with $N^{\,n}(k) = \sum_{j<k} f_j^{\,n}\, V_j'^{\,n} \Delta r_j^{\,n}$ *the cumulative particle count enclosed by old-grid face* k*, and*

$$\mathcal{N}^{\,n}(\hat{\psi}) \;=\; \mathrm{interp}\Big(\hat{\psi};\; \hat{\psi}^{\mathrm{mapped}}(r_f^{\,n}),\, N^{\,n}\Big) \tag{3.3.9}$$

its interpolant as a function of the mapped flux label. In particular, $\Delta N_{\mathrm{lost}} = 0$*, and the scheme is exactly particle-conserving, whenever the new grid's outer face coincides with the tracer-mapped image of the old domain edge, Equation* (3.3.17)*; this holds in particular for the identity map on an unchanged grid.*

Since (p, ξ) *were fixed but arbitrary, this conservation property, including the identical boundary term* ΔN_{lost}*, holds independently at every point of the momentum grid, and therefore also for any moment obtained by integrating over* (p, ξ) *– in particular the density* $n_{\mathrm{re}} = \int f_{\mathrm{re}}\, \mathrm{d}p\, \mathrm{d}\xi$*.*

Proof. To lighten notation, the momentum-space arguments are suppressed throughout, consistent with the statement of the lemma.

Write the cumulative enclosed particle count on the old grid's cell faces at timestep n as

$$N^{\,n}(k) \;=\; \sum_{j<k} f_j^{\,n}\, V_j'^{\,n} \Delta r_j^{\,n}, \qquad N^{\,n}(0) = 0. \tag{3.3.10}$$

This is the cumulative particle count enclosed by old-grid face k, defined at the flux label $\hat{\psi}(r_{f,k}^{\,n})$ (step 6 of Algorithm 3). Then, steps 2–3 push each of these face labels through the tracer map to $\hat{\psi}^{\mathrm{mapped}}(r_{f,k}^{\,n})$, without altering the value $N^{\,n}(k)$ attached to it: the map only relocates where that enclosed count sits in $\hat{\psi}$-space.

Step 6 reinterpolates this same cumulative array at the new grid's face labels $\hat{\psi}(r_{f,i}^{\,n+1})$, $i = 0, \ldots, M$, using the interpolant of Equation (3.3.9):

$$N^{\,n+1}(i) \;=\; \mathcal{N}^{\,n}(\hat{\psi}(r_{f,i}^{\,n+1})), \tag{3.3.11}$$

and step 7 differentiates and divides by the new cell volumes:

$$f_i^{\,n+1} \;=\; \frac{N^{\,n+1}(i{+}1) - N^{\,n+1}(i)}{V_i'^{\,n+1} \Delta r_i^{\,n+1}}. \tag{3.3.12}$$

Summing $f_i^{\,n+1} V_i'^{\,n+1} \Delta r_i^{\,n+1}$ over all M new cells telescopes exactly, since it is a sum of consecutive differences of $N^{\,n+1}$:

$$\sum_{i=0}^{M-1} f_i^{\,n+1}\, V_i'^{\,n+1} \Delta r_i^{\,n+1} = N^{\,n+1}(M) - N^{\,n+1}(0), \tag{3.3.13}$$

which holds regardless of grid resolution, spacing, or the detailed shape of $\hat{\psi}^{\mathrm{mapped}}$.

There are now a few cases regarding the particle count at the boundary of the domain. First, for the inner boundary, as $\hat{\psi}$ denotes the normalized poloidal flux, it follows by definition that $\hat{\psi}(r_{f,0}^{\,n}) = \hat{\psi}(r_{f,0}^{\,n+1}) = 0$ on both equilibria, regardless of the physical shift of the magnetic axis, since the innermost face always sits at the axis by construction of the normalized coordinate. Hence

$$N^{\,n+1}(0) = N^{\,n}(0) = 0 \tag{3.3.14}$$

holds exactly and requires no assumption about the tracer map or the grids: the interpolant is evaluated at $\hat{\psi} = 0$ on both sides, which is one of its own sample points by definition.

No analogous guarantee holds at the outer face: since DREAM's radial domain is not itself anchored to a fixed normalized flux value, the old and new grids' outer faces $r_{f,K}^{n}$ and $r_{f,M}^{n+1}$ are, in general, different physical locations, and there is no reason for $\hat{\psi}(r_{f,M}^{n+1})$ and $\hat{\psi}^{\mathrm{mapped}}(r_{f,K}^{n})$ to coincide. In this generic case,

$$N^{n+1}(M) = \mathcal{N}^{n}(\hat{\psi}(r_{f,M}^{n+1})) \neq N^{n}(K), \tag{3.3.15}$$

and (3.3.13), together with $N^{n+1}(0) = 0$ from the inner boundary, gives

$$\sum_{i=0}^{M-1} f_i^{n+1} V_i'^{n+1} \Delta r_i^{n+1} = \mathcal{N}^{n}(\hat{\psi}(r_{f,M}^{n+1})) = N^{n}(K) - \Delta N_{\mathrm{lost}}, \tag{3.3.16}$$

with ΔN_{lost} as defined in Equation (3.3.8), establishing Equation (3.3.7).

This boundary term is a small particle loss or gain created through the mismatch between the new grid's outer face and the tracer-mapped old domain. It actually reflects the evolution of the flux surfaces, as particles that occupied the region:

$$\hat{\psi} \in (\hat{\psi}(r_{f,M}^{n+1}),\ \hat{\psi}^{\mathrm{mapped}}(r_{f,K}^{n}))$$

(or vice versa) at step n have been carried outside, or were never included inside, the new grid's domain, and are correspondingly excluded from, or introduced into, the tracked particle count. Physically, this reflects the scraping-off of outer flux surfaces, when a vertically moving RE beam moves into the walls.

If the outer faces happen to satisfy

$$\hat{\psi}(r_{f,M}^{n+1}) = \hat{\psi}^{\mathrm{mapped}}(r_{f,K}^{n}), \tag{3.3.17}$$

e.g. because the new grid's outer face is deliberately placed at the tracer-mapped image of the old domain edge, then $\Delta N_{\mathrm{lost}} = 0$, and the total particle count is conserved to machine precision. In particular, the identity map $\hat{\psi}^{\mathrm{mapped}} = \hat{\psi}$ on an unchanged grid satisfies Equation (3.3.17) face-by-face, i.e. $\hat{\psi}(r_{f,i}^{n+1}) = \hat{\psi}^{\mathrm{mapped}}(r_{f,i}^{n})$ for every i, so the interpolation in step 6 returns N^{n} at all of its own nodes exactly, giving $N^{n+1} = N^{n}$ and $f^{n+1} = f^{n}$ cell by cell – recovering the stated identity-transform limit. □

3.3.4 Projecting the runaway electron current onto the JOREK computational grid

Since $j_{\mathrm{RE,DREAM}}(r)$ and $B_{\mathrm{min}}(r)$ are computed on DREAM's own radial grid, they are first interpolated onto the fixed normalized flux grid $\{\hat{\psi}_k\}_{k=1}^{n_\psi}$ that JOREK originally sent to DREAM (Section 3.2), before being sent back through the coupling. Once JOREK receives $j_{\mathrm{RE,DREAM}}(\hat{\psi})$ and $B_{\mathrm{min}}(\hat{\psi})$ from DREAM, it must evaluate the runaway current at arbitrary (R, Z) points dictated by its own finite element mesh. Since the current density computed at a given flux surface label in DREAM corresponds to the one at the point of lowest magnetic field strength, B_{min}, the runaway current for an arbitrary point (R, Z) in JOREK can be expressed as:

$$j_{\mathrm{r}}(R, Z) = \frac{B(R, Z)}{B_{\mathrm{min}}(\hat{\psi}(R, Z))}\, j_{\mathrm{RE,DREAM}}(\hat{\psi}(R, Z)). \tag{3.3.18}$$

Both $j_{\mathrm{RE,DREAM}}$ and B_{min} are stored on the same fixed grid $\{\hat{\psi}_k\}_{k=1}^{n_\psi}$ used to build the equilibrium sent to DREAM, so no re-interpolation onto a different radial grid is required.

At an arbitrary point with normalised flux $\hat{\psi}^*$, both quantities are evaluated by linear interpolation:

$$j_{\mathrm{RE,DREAM}}(\hat{\psi}^*) = (1-\alpha)\, j_{\mathrm{RE,DREAM}}(\hat{\psi}_k) + \alpha\, j_{\mathrm{RE,DREAM}}(\hat{\psi}_{k+1}), \tag{3.3.19}$$

$$B_{\min}(\hat{\psi}^*) = (1-\alpha)\, B_{\min}(\hat{\psi}_k) + \alpha\, B_{\min}(\hat{\psi}_{k+1}), \tag{3.3.20}$$

where k is the index such that $\hat{\psi}_k \leq \hat{\psi}^* < \hat{\psi}_{k+1}$ and $\alpha = (\hat{\psi}^* - \hat{\psi}_k)/(\hat{\psi}_{k+1} - \hat{\psi}_k)$. Outside the last closed flux surface ($\hat{\psi} \geq 1$), the RE current is set to zero.

It is worth noting that DREAM operates in SI units, while JOREK uses its own system of normalized units. The runaway current received from DREAM $j_{\mathrm{RE,DREAM}}$ is in A/m^2 and is then converted to JOREK units by a multiplicate factor of $\mu_0 R$, to obtain the normalized toroidal current density, it needs to additionally be multiplied by a factor of $\frac{F_0}{B\,R}$ [14], where R is the the local major radius and B is the magnitude of the **B**-field at the evaluation point. By doing this conversion, j_{r} in Equation (3.3.18) corresponds to the normalized toroidal current density introduced in Section 2.2.

In practice, Equation (3.3.18), including the unit conversion described above, is evaluated directly at the Gaussian quadrature points used to compute the weak-form integrals within each element (Section 2.2.2), where R, B and $\hat{\psi}$ are already available. The resulting runaway current density then enters the induction equation (2.2.9).

Chapter 4

Numerical validation of the coupled model and first results

The goal of the JOREK-DREAM coupling is to alleviate the computational cost of kinetic runaway-electron simulations in JOREK over the very long CQ and plateau time scales, in scenarios where 3D MHD activity does not play a significant role. Since the duration of the CQ and plateau phases increases with machine size, it makes sense to test the coupling on a scenario representative of a large tokamak. For this reason, the coupled model is validated against previously conducted 2D JOREK RE fluid simulations of the EU-DEMO tokamak [64]. In this work, the validation is limited to the early CQ phase, up to the formation of the RE plateau. Since the benchmark itself uses a fluid RE model, it cannot be used to verify the evolution of the distribution function during the coupled simulation; instead, it targets two properties of the coupling:

- Similar evolution trends of the plasma quantities: total plasma current, RE current, parallel electric field, temperature.
- Correct tracking of the evolution of the magnetic flux surfaces during the vertical motion of the plasma.

Section 4.1 describes the benchmark scenario and the setup used for the coupled JOREK-DREAM simulations. Section 4.2 then verifies the flux-surface tracker and the quasi-conservative remapping algorithm independently of the coupled physics. Finally, Section 4.3 examines the convergence of the coupled simulation with respect to the spatial and temporal resolution parameters and then compares the coupled model against the benchmark and presents the first physical results obtained with the coupling.

4.1 Benchmark scenario and simulation setup

4.1.1 Benchmark scenario

The benchmark scenario considered is a full disruption scenario of the EU-DEMO tokamak containing a VDE. It has been modeled using the JOREK reduced MHD model coupled to the RE fluid model [65, 66] and the CARIDDI code [36] to take care of the free-boundary and resistive wall physics. The initial equilibrium is reconstructed from an EQDSK equilibrium file for a EU-DEMO scenario (`Equil_DEMO_delta95_0d25_SOB_sc`), with toroidal field $B_\phi = 4.392\,\mathrm{T}$, plasma current $I_p = 18.79\,\mathrm{MA}$, and central pressure $p_0 = 9.56 \times 10^5\,\mathrm{Pa}$. The corresponding geometric major radius is $R_{\mathrm{geo}} = 8.55\,\mathrm{m}$ and

toroidal flux function $F_0 = RB_\phi = -37.75\,\mathrm{T\,m}$. The poloidal cross-section is resolved with $n_{\mathrm{radial}} = 100$ radial and $n_{\mathrm{pol}} = 150$ poloidal grid points.

The plasma interaction with the resistive vacuum vessel and 15 poloidal-field coils is captured via the self-consistent eddy-current model provided by the JOREK-CARIDDI coupling. The VDE is triggered by an artificial perturbation applied to one in-vessel control coil: its current is ramped by $-1 \times 10^4\,\mathrm{A}$ starting at $t = 0$, growing over the timescale of the pre-disruption free-boundary evolution phase, which provides an initial perturbation for the exponentially growing vertical motion of the plasma into the wall.

In this reference run, runaway electron generation is modeled entirely within JOREK's own fluid RE model [65, 66] and with a seed RE current profile prescribed as a scaled-down copy of the total current profile at an initial fraction of 10^{-3} of I_p, and avalanche (secondary) generation enabled throughout.

The reference simulation is divided into six phases taking care of the different phases of the disruption. Steps 1–1850 (`in_EQ` through `in_IMP`) construct a physically consistent pre-CQ plasma state (i.e., they reflect a simplified axisymmetric treatment of the thermal quench dynamics used here to concentrate on the successive phases) and are not themselves part of the compared results. The reference JOREK simulation and both DREAM-coupled configurations restart from the same JOREK state at step 1850 to share identical initial conditions and are carried out over the initial phase of the current quench up to formation of the RE plateau. Studies of the full CQ including the entire plateau phase are left for future work. The normalized time step in the reference simulation is set to $t_{\mathrm{norm}} \approx 0.61\,\mu\mathrm{s}$ and all timesteps in Table 4.1 are given as multiples of this quantity. Note that the JOREK fluid time step is not held fixed at this value in the coupled runs presented in this thesis: as detailed in Section 4.3, the coupling time step is instead varied as part of the convergence study, and the JOREK step ranges listed in Table 4.1 therefore only apply to the reference simulation used to construct and label the benchmark scenario, not to the coupled runs analyzed in the remainder of this chapter.

Phase	JOREK step range	Physical duration
`in_EQ`	1–1200	$\approx 1.26\,\mathrm{ms}$
`in_TQ1`	1201–1650	$\approx 3.25\,\mathrm{ms}$
`in_TQ2`	1651–1800	$\approx 1.27\,\mathrm{ms}$
`in_IMP`	1801–1850	$\approx 0.07\,\mathrm{ms}$
`in_CQ1`	1851–5400	$\approx 61.93\,\mathrm{ms}$
`in_CQ2`	5401–11400	$\approx 110.23\,\mathrm{ms}$

Table 4.1: Summary of the phases of the benchmark scenario simulation. Only a portion of the `in_CQ1` phase, corresponding to the early current quench (step ≥1850) is analyzed in this thesis, while the full simulation of the CQ and RE plateau is left for future work. Step ranges and durations correspond to the reference JOREK simulation only; the coupled runs use a varying JOREK fluid time step (Section 4.3) and are therefore not indexed by the same step numbers.

At the start of the simulation, the radial profiles for the runaway electron density, the initial electron temperature and the initial parallel electric field in the benchmark scenario are visualized in Figure 4.1. These correspond to post TQ/early CQ profiles taken as starting point for the successive studies (see Figure 2.4).

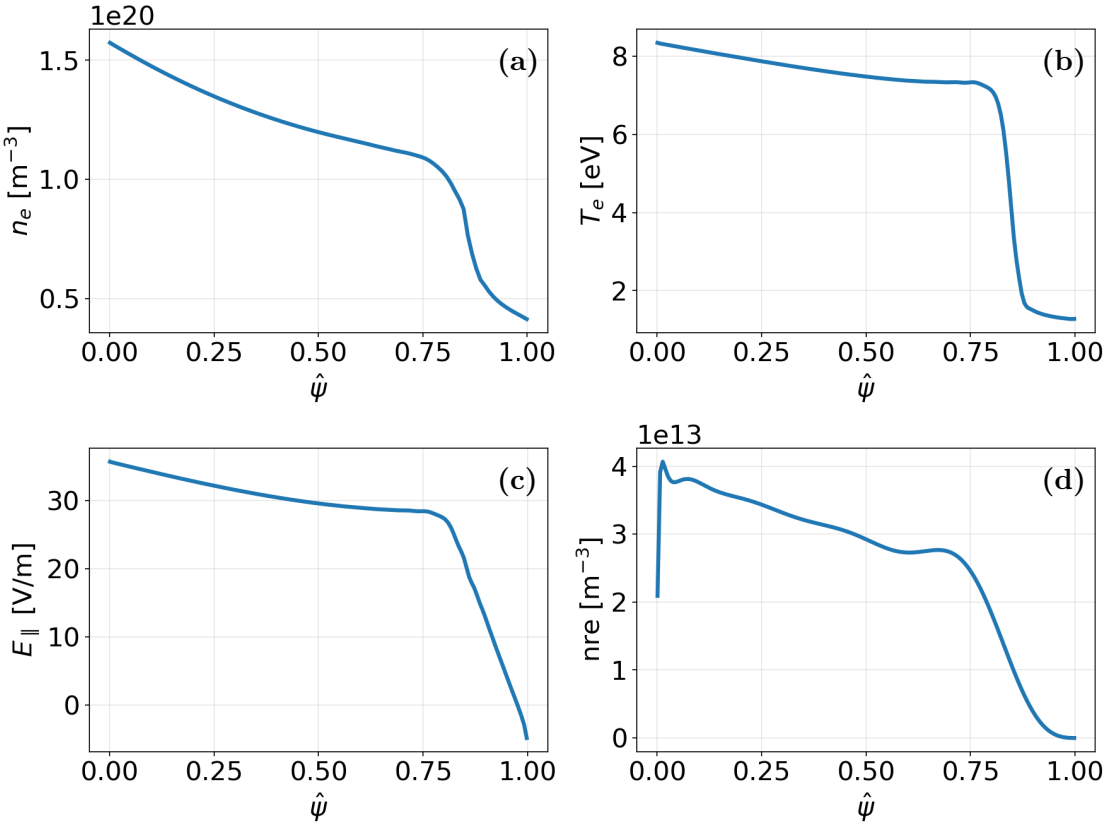


Figure 4.1: Initial profiles of (a) the electron density, (b) the electron temperature, (c) the parallel electric field, and (d) the runaway electron density in the benchmark scenario used as initial state for the studies.

4.1.2 Setup of the coupled simulations

At the first coupling step, the hot electron and runaway electron distribution functions, f_{hot} and f_{re}, are initialized as described below, since no prior DREAM output exists to restart from. At all subsequent steps, both distribution functions are instead re-initialized based on their state at the end of the previous DREAM simulation, taking into account the mapping from the previous flux surfaces to the new ones as described in Section 3.3.3.

The hot electron distribution is initialized as an isotropic Maxwellian (thermal-equilibrium momentum distribution with no dependence on the pitch angle ξ_0), consistent with the prescribed electron temperature and density profiles at the first coupling time step $n_e(r) = n_0(r)$ and $T_e(r) = T_0(r)$ (shown in Figures 4.1(a) and 4.1(b)):

$$f_{\mathrm{hot}}(r,p) = \frac{n_0(r)}{\pi^{3/2}\, p_T(r)^3} \exp\!\left(-\frac{p^2}{p_T(r)^2}\right), \tag{4.1.1}$$

where the thermal momentum spread is defined as

$$p_T(r) = \sqrt{\frac{2T_0(r)}{m_e c^2}}\,. \tag{4.1.2}$$

The runaway population is instead seeded with a distribution shaped to emulate a hot-tail-generated population: a narrow bulk of electrons just above the critical runaway momentum, together with a smaller secondary population ("blob") further out in momentum space. The unnormalized shape function is

$$\tilde{f}_{\mathrm{re}}(r,p,\xi_0) = \left[\exp\!\left(-\frac{(p-p_{\mathrm{bulk}})^2}{2\sigma_{\mathrm{bulk}}^2}\right) + f_{\mathrm{blob}} \exp\!\left(-\frac{(p-p_{\mathrm{blob}})^2}{2\sigma_{\mathrm{blob}}^2}\right)\right] \exp\!\left(-\frac{\big(\xi_0 - \xi_{\mathrm{peak}}(r)\big)^2}{2\sigma_\xi^2}\right), \tag{4.1.3}$$

with the momentum-space parameters, expressed in units of the runaway grid spacing Δp_{re},

$$p_{\mathrm{bulk}} = 2.0\,\Delta p_{\mathrm{re}}, \qquad \sigma_{\mathrm{bulk}} = 1.5\,\Delta p_{\mathrm{re}}, \tag{4.1.4}$$

$$p_{\mathrm{blob}} = p_{\mathrm{bulk}} + 4.0\,\Delta p_{\mathrm{re}}, \qquad \sigma_{\mathrm{blob}} = 1.5\,\Delta p_{\mathrm{re}}, \tag{4.1.5}$$

a relative blob amplitude $f_{\mathrm{blob}} = 0.15$, and a pitch-angle spread $\sigma_\xi = 0.15$. The distribution is forward-peaked along the direction set by the local parallel electric field,

$$\xi_{\mathrm{peak}}(r) = \mathrm{sign}\,\big(E_\parallel(r)\big), \tag{4.1.6}$$

consistent with runaway electrons streaming nearly parallel to $\mathbf{B}$ in the direction of local acceleration.

The shape function (4.1.3) is then rescaled radially so that its density moment matches a prescribed target profile $n_{\mathrm{re}}^{\mathrm{target}}(r)$, which is given below. The zeroth moment of the shape function is evaluated numerically using DREAM's own momentum-space integration weights,

$$n_{\mathrm{shape}}(r) = \int_{-1}^{1} \mathrm{d}\xi_0 \int_0^{p_{\max}} \mathrm{d}p\; p^2\, \tilde{f}_{\mathrm{re}}(r,p,\xi_0), \tag{4.1.7}$$

and the initial runaway distribution is obtained as

$$f_{\mathrm{re}}(r,p,\xi_0) = \tilde{f}_{\mathrm{re}}(r,p,\xi_0)\, \frac{n_{\mathrm{re}}^{\mathrm{target}}(r)}{n_{\mathrm{shape}}(r)}, \qquad n_{\mathrm{re}}^{\mathrm{target}}(r) = \max\big(n_{\mathrm{re,ini}}(r),\, 0\big). \tag{4.1.8}$$

The initial runaway density profile $n_{\mathrm{re,ini}}(r)$ is chosen to be the one in JOREK at the first coupling step (shown in Figure 4.1(d)).

For the momentum grid resolutions, the values of $p_{\mathrm{hot}} = 2.5$ and $p_{\mathrm{re}} = 300.5$ are chosen (see Section 2.3.1 for their definition). The hottail grid is bi-uniform in momentum with increased resolution before $p_{\mathrm{sep}} = 0.07$ to resolve the thermal bulk correctly and it is bi-uniform in pitch angle with increased resolution before $\xi_{\mathrm{sep}} = 0.5$. A dedicated trapped–passing boundary-layer grid is additionally applied, with maximum pitch-angle spacing $\Delta\xi_{\mathrm{max}} = 2 \times 10^{-2}$ and boundary-layer width 10^{-4}, in order to resolve the trapped–passing boundary in the hot population. In the runaway grid, only the pitch-angle coordinate is biuniformly refined, again split at $\xi_{\mathrm{sep}} = 0.5$, since the runaway population does not require the same momentum-space refinement near the thermal bulk. Both distributions are advanced using an upwind interpolation scheme for the advection terms (together with the corresponding upwind Jacobian). The generation of new runaway electrons is modelled through the kinetic avalanche mechanism, with a momentum cutoff for the avalanche source of $p_{\mathrm{cut}} = 0.01\, mc$.

The physical timestep Δt to be covered by each DREAM simulation is subdivided into N_t internal DREAM substeps, $N_t = \max(1, \lceil \Delta t / \Delta t_{\mathrm{target}} \rceil)$ with $\Delta t_{\mathrm{target}} = 10^{-7}$ s.

The ion background is set as described in Section 2.3.1, specifically as in Equation (2.3.12).

On the JOREK side of the coupling, the values of $n_\psi = 100$ and $n_\theta = 128$ are chosen for the computation of the flux surface geometry and for the flux surface mapping (see Section 3.3.2).

4.2 Verification of the numerical algorithms of the coupling

4.2.1 Verification of the flux-surface tracking and particle tracer

Algorithm 2 has the purpose of tracking the evolution of the flux surfaces in JOREK in between coupling steps. Two fundamental properties are verified here: that the tracer correctly converges to the identity map when the tracing time interval is reduced to zero, and that it correctly converges with regards to the pushing step size.

Convergence to the identity map

When the tracing time interval is reduced, it is expected that the tracing algorithm converges to the identity mapping, which means that for each flux surface $s = 1, ..., n_\psi$ it needs to be checked that:

$$\lim_{\Delta t \to 0} \hat{\psi}_s^{\mathrm{mapped}} = \hat{\psi}_s. \tag{4.2.1}$$

This is verified by calling the tracing algorithm for smaller and smaller time intervals, leading to $\Delta t \to 0$. It is assessed by measuring the L_2 norm of the deviation from the identity map over the n_ψ traced flux surfaces,

$$\left\| \hat{\psi}^{\mathrm{mapped}} - \hat{\psi} \right\|_2 = \sqrt{\frac{1}{n_\psi} \sum_{s=1}^{n_\psi} \left(\hat{\psi}_s^{\mathrm{mapped}} - \hat{\psi}_s \right)^2}. \tag{4.2.2}$$

To do this, the tracing time intervals are constructed as:

$$[t_n, t_n + x \cdot (t_{n+1} - t_n)], \tag{4.2.3}$$

where x is the shrinkage factor $0, \frac{1}{2}, \frac{1}{4}$ which reduces the length of the interval. The number of selected flux surfaces is $n_\psi = 15$. The selected fluid time step is taken to be the second one of the coupling (i.e. time step 1852). The length of this time step and thus of the first interval is of $6.124 \cdot 10^{-8}$ s. The particles are pushed with a timestep of 10^{-11} s.

The results obtained are shown in Figure 4.2.

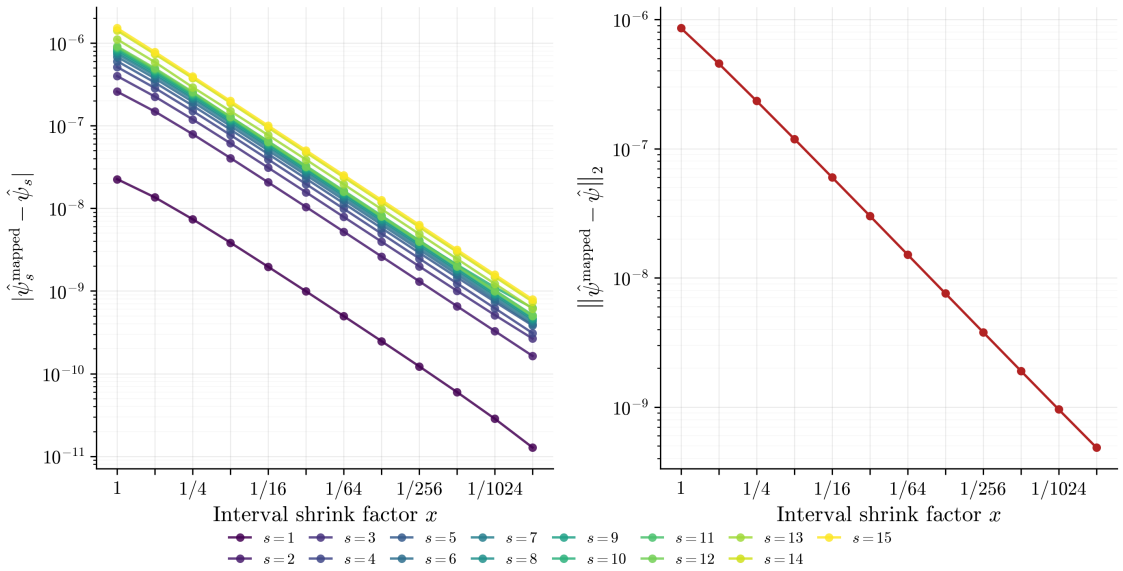


Figure 4.2: Convergence of the flux-surface tracer toward the identity mapping as the traced time interval $\Delta t \to 0$.

As observed, the error diminishes with the length of the interval, indicating that the mapping approaches the identity mapping, as expected.

Convergence of the guiding-centre pusher

To verify that the internal particle-push step size Δ_t used in the flux-surface tracing routine is sufficiently small, a convergence study was performed in which the full coupling interval was traced repeatedly while Δ_t was halved successively from 1×10^{-7} s down to 1.5625×10^{-9} s. For each Δ_t, the deviation of the resulting mapped flux-surface coordinate $\hat{\psi}_s^{\text{mapped}}$ from a reference trace computed at the production step size ($\Delta_t = 10^{-9}$ s) was recorded for each of the fifteen traced flux surfaces. The same JOREK timestep as for the identity map convergence test is selected here. Figure 4.3 shows this deviation as a function of the particle pushing step size Δ_t. It is observed that for values below 1.2×10^{-8} s, the deviation from the production run is constant, indicating that reducing the pushing time step further provides no improvement.

4.2.2 Verification of the conservative remapping scheme

The verification of the quasi-conservative property of the remapping scheme was established in Section 3.3.3. Now attention is turned to the numerical verification of this property.

The remap algorithm 3 is tested on a manufactured distribution function evaluated at a fixed, arbitrary momentum-grid point (p, ξ) and on synthetic flux-label maps, independent of any JOREK or DREAM output, so that the expected result is known analytically in

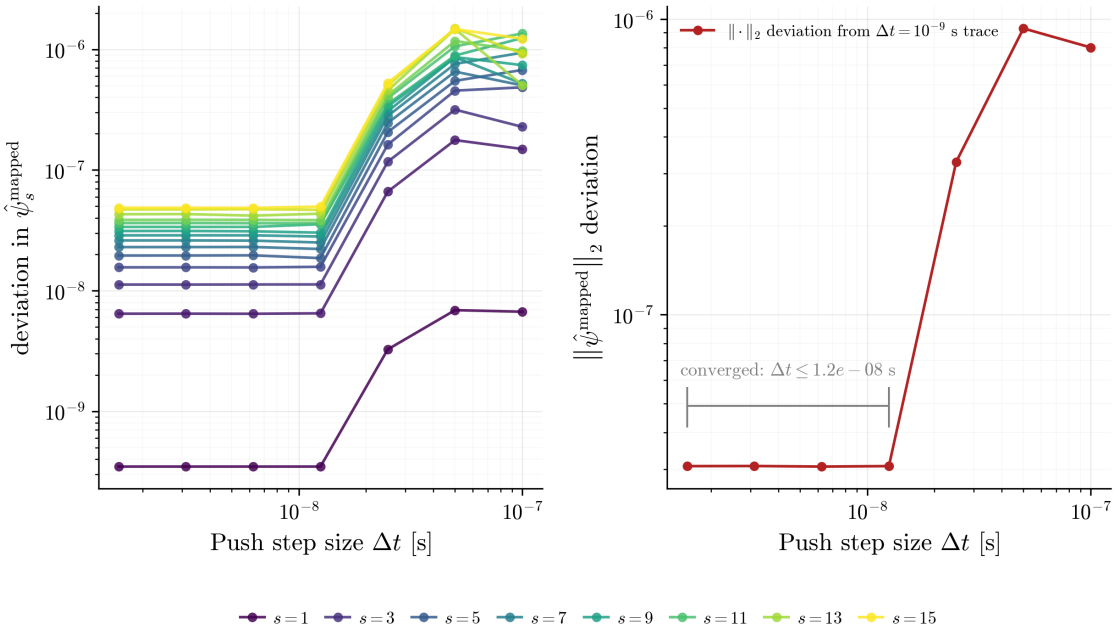


Figure 4.3: Convergence of the flux-surface tracer as the pushing timestep is diminished.

every case; as established in Lemma 1, conservation at one such fixed (p, ξ) is representative of the full momentum grid. The manufactured radial grid uses a smooth, strictly positive volume-Jacobian factor

$$V'(r) = 2\pi r\,(1 + 0.1\sin(3r)), \tag{4.2.4}$$

chosen to mimic DREAM's geometric Jacobian without reducing to the trivial cylindrical case, and a smooth, strictly monotonic normalized flux label

$$\hat{\psi}(r) = \min\left\{\max\left[\left(\frac{r}{a}\right)^{1.3},\ 10^{-3}\right],\ 0.999\right\}, \qquad \hat{\psi}(r) \in \left[10^{-3},\, 0.999\right], \tag{4.2.5}$$

clipped away from 0 and 1 to avoid degenerate interpolation endpoints. The manufactured distribution value at the fixed (p, ξ) is a smooth, core-peaked, strictly positive radial profile,

$$f(r; p, \xi) = 10^{15}\exp\left[-\left(\frac{r}{0.3\,a}\right)^2\right] + 10^{12}\ \mathrm{m}^{-3}, \tag{4.2.6}$$

i.e. a Gaussian of characteristic width $0.3\,a$ centred on the magnetic axis, with a small uniform floor added so the profile never vanishes at large r. The value of $a = 1$ is chosen for simplicity. None of V', $\hat{\psi}$, or $f(r; p, \xi)$ corresponds to an actual JOREK/DREAM equilibrium, as mentioned.

The considered test cases are the following:

- **Identity map, same grid:** $\psi_{n,\text{mapped}} = \psi_n$, old and new grids identical.
- **Identity map, different (finer) new grid:** tests that the conservation is independent of a change in radial resolution across the remap.
- **Interior deformation:** a smooth, strictly monotonic but nontrivial flux-label remapping ($\psi_{n,\text{mapped}} = \psi_n^{1.7}$, compressing the core and expanding the edge) with matching domain endpoints; tests conservation under a non-identity, physically-representative displacement.

- **Boundary loss:** the outermost flux labels are mapped beyond their nominal domain, deliberately displacing material past the new grid's outer boundary; the numerical result is compared against the analytically expected loss, rather than against the undisturbed total.

Table 4.2 reports the relative error between the remapped total and the expected total for each case.

Test case	Relative conservation error
Identity map, same grid	-2×10^{-16}
Identity map, different (finer) new grid	-2×10^{-16}
Interior deformation	-2×10^{-16}
Boundary loss vs. analytically predicted loss	0

Table 4.2: Conservation error of the remap algorithm on a manufactured distribution value $f(r; p, \xi)$ at a fixed momentum-grid point and synthetic displacement maps. The error is independent of grid resolution and of the strength of the interior deformation, confirming that the conservation property derived earlier holds exactly in the implementation; the boundary-loss case confirms that when the tracer map displaces material beyond the new domain, the resulting loss matches the analytically expected value exactly.

This verifies conservation of the total particle count, at a representative momentum-grid point, across a coupling timestep, and by Lemma 1, of the momentum-integrated density n_{re} as well.

4.3 Validation of the coupled model against the benchmark

After verifying the new numerical algorithms implemented for the purpose of the coupling, the spatial and temporal convergence of the coupled simulation is checked in Section 4.3.1 and then the comparison of the coupled model against the benchmark scenario is presented in Section 4.3.2.

4.3.1 Convergence of the coupled simulation

To verify the correctness of the implementation, it is necessary to check that a change in the spatial and temporal resolutions does not affect the results quantitatively. Since no exact solution of the coupled problem is available, the convergence of the discretization cannot be measured directly against a reference solution. Instead, a self-convergence test is performed: the spatial and temporal resolutions are increased successively, and it is checked that the results approach each other, with the run at the highest resolution taken as the reference. The differences with respect to this reference are referred to as errors in the following, although they only approximate the true discretization error, and only if the reference run is itself sufficiently well resolved. Three types of resolution are considered. First, the radial resolution N_r, the number of radial grid points on which DREAM's flux-surface-averaged quantities are represented, is varied. Second, the momentum resolution, given by the pair $(N_{p,\mathrm{hot}}, N_{p,\mathrm{re}})$ setting the number of grid points in momentum space for the hot and runaway electron populations respectively, is varied. Finally, the temporal resolution, set by the JOREK-DREAM coupling time step (and thus the length of the JOREK time step), is varied.

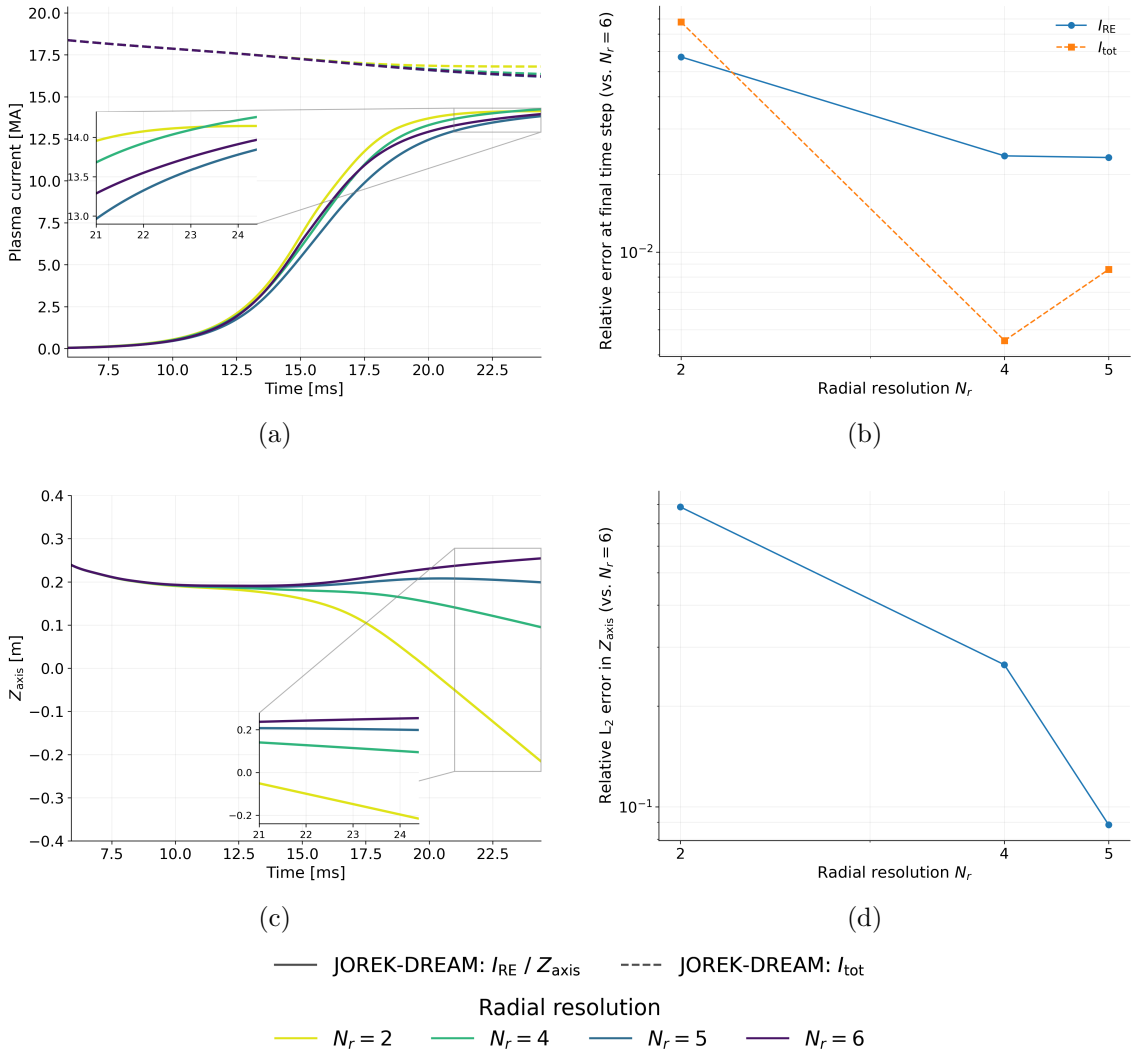


Figure 4.4: Convergence of the coupled simulation with respect to the radial resolution N_r: (a) total plasma and RE current, (b) relative L_2 error in the RE current, (c) magnetic axis vertical position Z_{axis}, (d) relative L_2 error in Z_{axis}.

Radial convergence

For the scan in the radial resolution, a constant fluid time step of length 6.1μs ($10t_{\text{norm}}$) is chosen and the momentum space resolution is set to $N_{\text{p,hot}} = 150$ and $N_{\text{p,re}} = 120$. The number of radial grid points in the DREAM simulations is varied according to $N_r = [2, 4, 5, 6]$. Due to the bad scaling of the computation time with the number of radial grid points, running with over $N_r = 6$ was found to be prohibitively expensive (while optimization of DREAM is being addressed in parallel activities independent from this thesis project). The results are shown in Figure 4.4.

The current curves in Figure 4.4a show that the total plasma current I_{tot} shows only a very weak sensitivity to the radial resolution, with all four curves overlapping closely throughout the simulated interval. The runaway current I_{RE} follow similar trends across N_r and converge to a common value at the end of the simulated time range as N_r increases, with the coarsest resolution, $N_r = 2$, deviating most from the others. This is confirmed in Figure 4.4b where the relative error in I_{RE}, compared to the $N_r = 6$ case, at the final time step decreases from $N_r = 2$ to $N_r = 4$ and then plateaus around the same level for $N_r = 5$, while the error in I_{tot} decreases sharply between $N_r = 2$ and $N_r = 4$, reaching its minimum there before increasing again slightly at $N_r = 5$. In both cases the errors remain below a few percent already from $N_r = 4$ onward. Exact numerical convergence checks with convergence order analysis are left for future work, since this will require the ability to increase the resolution in DREAM to substantially higher values.

The evolution of the magnetic axis position, shown in Figure 4.4c, is considerably more sensitive to the radial resolution than the current dynamics. While all resolutions agree closely up to around $t \approx 12$ ms, the coarsest case, $N_r = 2$, diverges strongly at later times, with Z_{axis} dropping to roughly -0.3 m by the end of the simulated window. The remaining resolutions, $N_r = 4$ and $N_r = 5$, stay closer to the reference $N_r = 6$ case but still show a visible, resolution-dependent spread over time. This is reflected in Figure 4.4d, where the relative L_2 error in Z_{axis} decreases monotonically with increasing N_r, dropping by roughly one order of magnitude between $N_r = 2$ and $N_r = 5$. Overall, the radial resolution scan indicates that while the plasma and runaway currents agree reasonably well at moderate resolution, the vertical position of the magnetic axis requires a higher radial resolution to be captured accurately. Simulations using higher values of N_r will be necessary to accurately predict the effect of the radial resolution on the runaway current and vertical displacement dynamics of the plasma.

Momentum space convergence

For the scan in the momentum grid resolution, a constant fluid time step of length 12.2 μs ($20t_{\text{norm}}$) is chosen and the radial resolution is set to $N_r = 6$. The momentum grid resolution is varied from $(N_{\text{p,hot}}, N_{\text{p,re}}) = (50, 40)$ to $(N_{\text{p,hot}}, N_{\text{p,re}}) = (300, 220)$. The results are shown in Figure 4.5.

The current curves in Figure 4.5a show that the total plasma current I_{tot} is again largely insensitive to the momentum-grid resolution, with all curves nearly overlapping and only a small residual spread visible in the inset at late times. The runaway current I_{RE}, on the other hand, shows a clear and monotonic trend: as the momentum resolution increases from $(N_{p,\text{hot}}, N_{p,\text{re}}) = (50, 40)$ to $(300, 220)$, the RE current rise shifts to later times and, as a result, the peak current observed in the simulated time window decreases, while the higher resolution cases lie closer together. This is confirmed in Figure 4.5b: the relative L_2 error in I_{RE}, evaluated against the finest resolution $(N_{p,\text{hot}}, N_{p,\text{re}}) = (300, 220)$, decreases monotonically with increasing momentum resolution, from over 50% at the coars-

est resolution to roughly 20% at $(150, 120)$, without yet showing clear evidence of having converged at the resolutions tested.

The evolution of the magnetic axis position, shown in Figure 4.5c, is less sensitive to the momentum resolution than the RE current: all curves stay close together for most of the simulated interval, with the coarsest resolution, $(50, 40)$, again showing the largest deviation, dipping visibly below the other curves at late times as seen in the inset. This is also seen in Figure 4.5d, where the relative L_2 error in Z_{axis} decreases sharply between $N_{p,\text{hot}} = 50$ and $N_{p,\text{hot}} = 150$, reaching a minimum around $N_{p,\text{hot}} = 150$, before increasing again at $N_{p,\text{hot}} = 200$. This non-monotonic behaviour indicates that, unlike the radial resolution scan, the error in Z_{axis} does not decrease uniformly with momentum resolution over the range tested. Overall, the momentum resolution scan shows that the runaway current is markedly more sensitive to the momentum-grid resolution than the total plasma current or the axis position, and that convergence in I_{RE} has not yet been clearly achieved even at the highest resolution considered. This incomplete convergence will need additional fine-grained analysis in the future and potentially some algorithmic changes in the coupled codes.

Temporal convergence

For the temporal resolution convergence scan, the number of radial points is set to $N_r = 6$ and the momentum space resolution is set to $(N_{\text{p,hot}}, N_{\text{p,re}}) = (150, 120)$. All runs share the same evolution up to $t = 9.9$ ms, from which point onward the coupling time step is varied, since this is when the current avalanche, and thus the evolution of the electric field, becomes important. The coupling time step is varied according to $\Delta t \approx [3.1, 6.1, 12.2, 18.3]\,\mu\text{s}$, which corresponds to $\Delta t = [5, 10, 20, 30]\,t_{\text{norm}}$. The results are shown in Figure 4.6.

The current curves in Figure 4.6a show that the three largest coupling time steps, $\Delta t = 18.4$, 12.2 and $6.1\,\mu$s, agree closely with one another throughout the simulated interval, both in I_{tot} and in I_{RE}, with only a small residual spread visible in the inset at late times. The finest time step considered, $\Delta t = 3.1\,\mu$s, however, departs markedly from the other three: the runaway current rises earlier and reaches a higher value during the current-quench phase, rather than converging more closely to the other curves as would be expected from a genuine refinement of the coupling time step. This is at odds with the usual convergence behaviour, where a finer resolution is expected to move progressively closer to, not further from, the other runs. The relative L_2 error in I_{RE} shown in Figure 4.6b, measured against the $\Delta t = 3.1\,\mu$s run, does decrease monotonically as Δt is reduced towards this value, but the errors themselves remain small throughout, only ranging between roughly 2.6% and 3.2%, which understates the qualitative disagreement visible in Figure 4.6a.

This behaviour indicates that the coupled model has not yet reached a clean, monotonic convergence regime with respect to the coupling time step indicating a potential problem in the co-evolution of the coupled system by the two codes, that will require further analysis and improvements before the model is ready for production applications. A more thorough investigation of this trend, for instance by extending the scan to even smaller time steps, or examining the DREAM-side substepping in more detail, is left for future work and falls outside the scope of this thesis. For this reason, the corresponding convergence check of the magnetic axis position Z_{axis} with respect to the coupling time step is not included here, since a meaningful Z_{axis} comparison would first require the underlying current dynamics to be well converged.

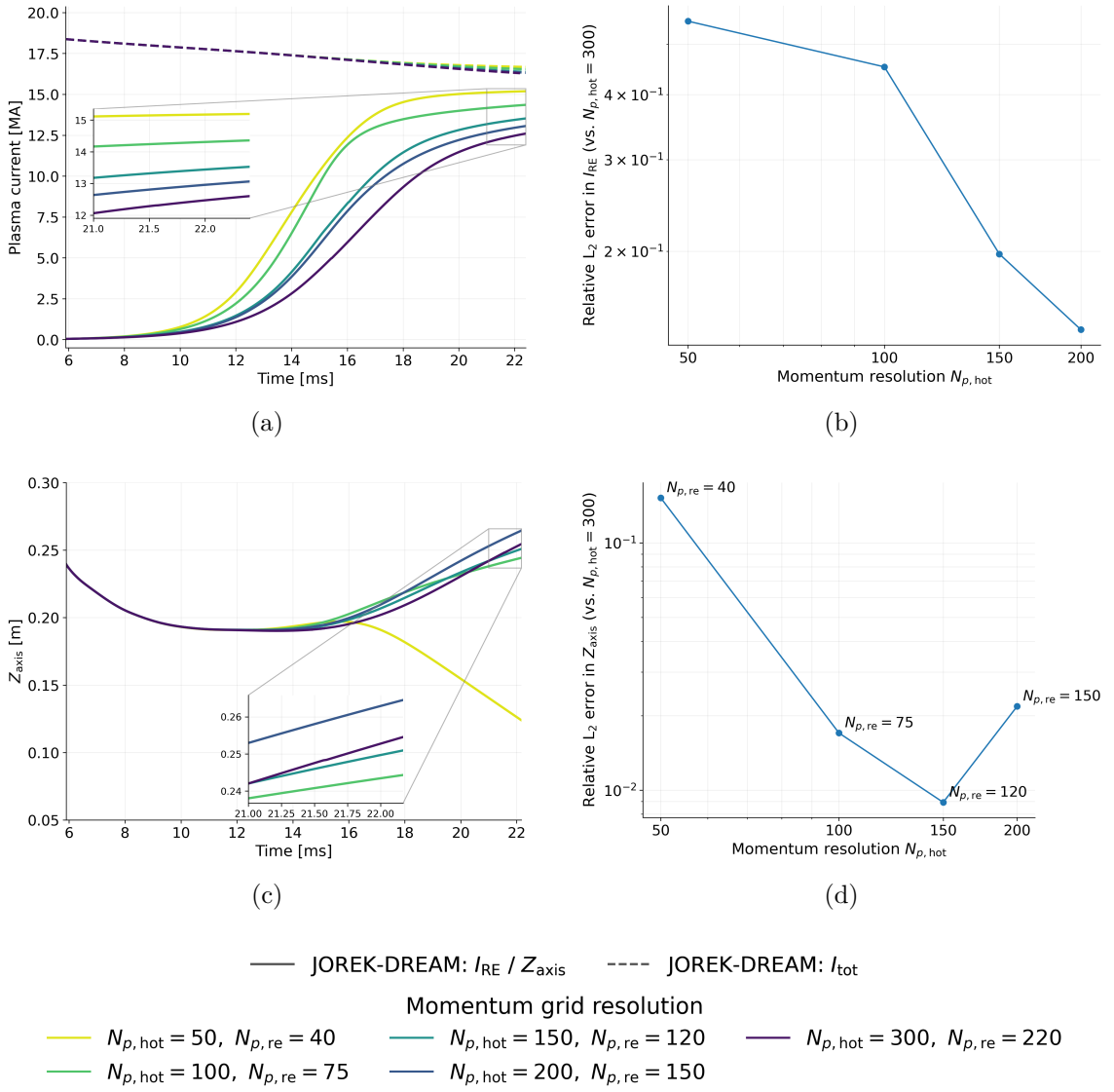


Figure 4.5: Convergence of the coupled simulation with respect to the momentum grid resolution ($N_{p,hot}, N_{p,re}$): (a) total plasma and RE current, (b) relative L_2 error in the RE current, (c) magnetic axis vertical position Z_{axis}, (d) relative L_2 error in Z_{axis}.

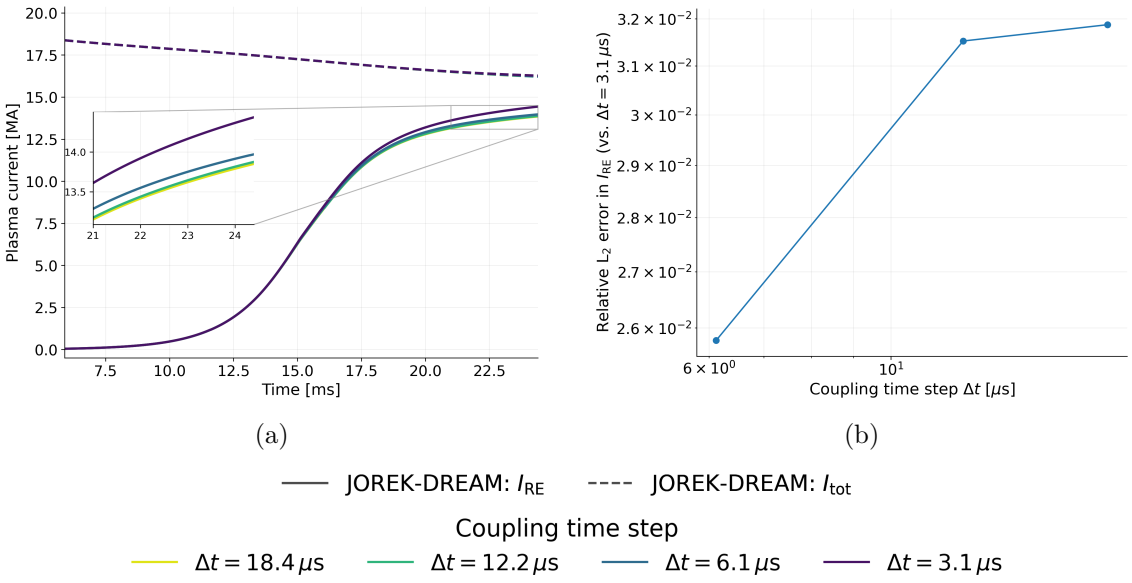


Figure 4.6: Convergence of the coupled simulation with respect to the coupling time step Δ_t: (a) total plasma and RE current, (b) relative L_2 error in the RE current.

4.3.2 Comparison of the coupled model against the benchmark and first results

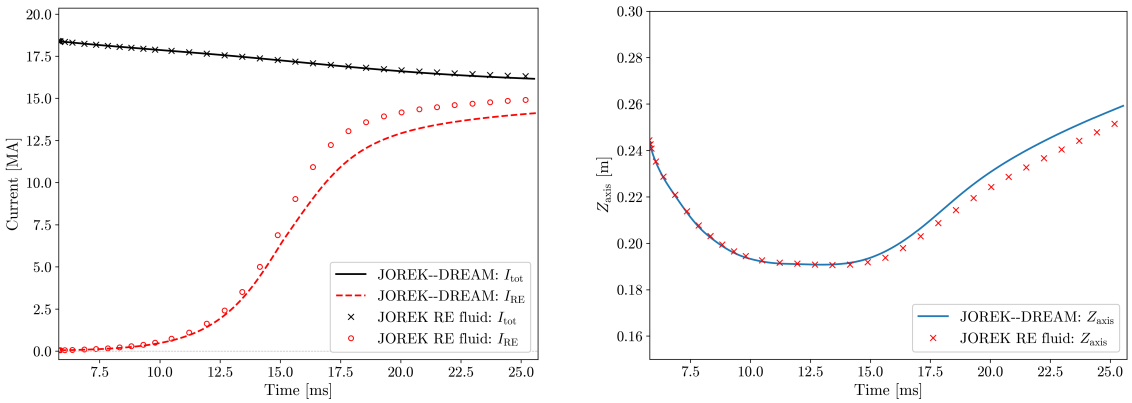
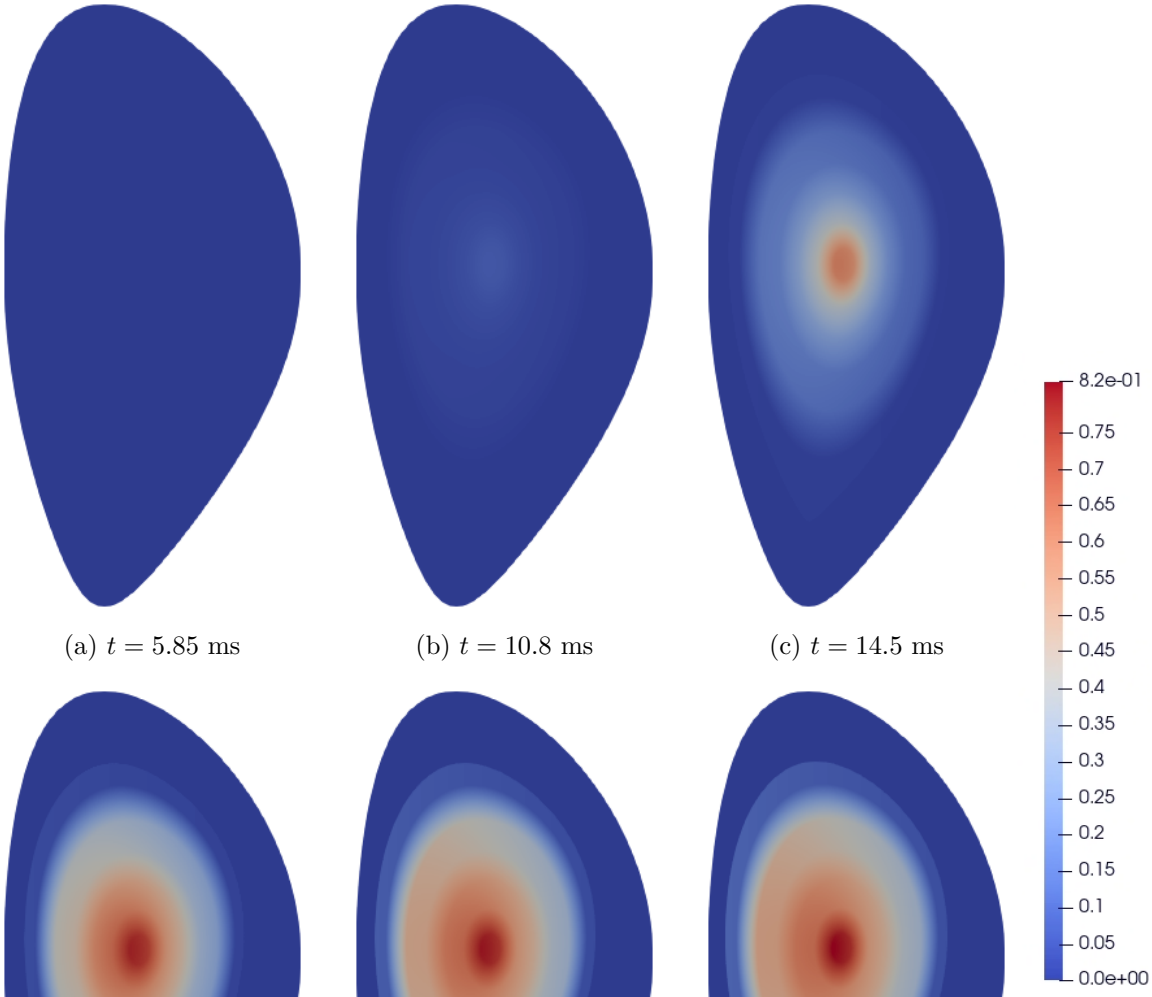


Figure 4.7: (a) Evolution of the total plasma and the runaway currents over time for the coupled model and the benchmark scenario. (b) Evolution of the z-axis position over time for the coupled model and the benchmark scenario.

After assessing the convergence properties of the coupled model, attention is turned to its comparison against the benchmark scenario and the first results regarding the kinetic distribution of the runaway population throughout the current quench.

The coupled simulation used for this analysis was run with $N_r = 6$ radial points in DREAM, momentum space resolution $(N_{\mathrm{p,hot}}, N_{\mathrm{p,re}}) = (150, 120)$ and with a coupling time step of $\Delta_t = 6.1\mu$s after $t = 9.9$ ms.

Figure 4.7 compares the coupled kinetic model against the benchmark scenario. Fig-



(a) $t = 5.85$ ms (b) $t = 10.8$ ms (c) $t = 14.5$ ms

(d) $t = 18.2$ ms (e) $t = 21.9$ ms (f) $t = 25.5$ ms

Figure 4.8: Poloidal projection of the runaway electron current density j_{RE} evolution over the course of the coupled simulation. The values of the colorbar are in MA/m^2.

ure 4.7a shows the evolution of the total plasma current and the RE current: the coupled model reproduces the slow decay of the total current, from approximately 18 MA to 15–16 MA over the simulated part of the current quench, in close agreement with the fluid benchmark. The RE current itself grows from its low seed value to roughly 14–15 MA by $t = 25$ ms, following a very similar trend as the fluid-based benchmark. The two models agree closely both before the onset of runaway generation and in the final plateau state. It is observed that the growth rate of the RE current is slightly different in both models, which is expected as the JOREK RE fluid model and JOREK-DREAM employ different physical models for the runaway physics.

Figure 4.7b shows the accompanying evolution of the vertical position of the plasma, Z_{axis}. Both models capture the same trend of the vertical displacement with a good general agreement. However, it is needed to simulate the full runaway plateau phase to accurately verify that the vertical displacement dynamics are correctly captured in the coupled model.

It should be emphasized that the benchmark does not represent an exact reference solution. It is also the result of a numerical model, in which the runaway electrons are described as a fluid. Differences between the two simulations can thus not be attributed to an error in either model, but to the different physical descriptions of the runaway population. The purpose of this comparison is therefore not to reproduce the benchmark exactly, but to verify that the coupled model yields a physically consistent evolution that agrees reasonably well with an established model. This is checked qualitatively, in the overall trends of the current and the vertical displacement, and quantitatively, in the magnitude and timing of the runaway current and the final plateau state.

The poloidal evolution of the RE current density j_{RE} over the same simulation is shown in Figure 4.8. No appreciable runaway current density is present at $t = 5.85$ ms, consistent with the current-quench-triggered avalanche not yet having set in. By $t = 10.8$ ms a weak, diffuse signal appears near the magnetic axis, and by $t = 14.5$ ms this has developed into a clearly localized peak, coincident with the onset of rapid RE current growth identified in Figure 4.7. Over the remainder of the simulation ($t = 18.2$–25.5 ms) the runaway beam continues to grow and eventually plateaus as expected from Figure 4.7a.

Finally, Figure 4.9 shows the evolution of the pitch-averaged electron distribution function at the point $r = 0$ in DREAM (core) throughout the coupled simulation. Over the entire considered time window, f_{hot} is sharply peaked at low momentum values, while f_{re} shows a flatter behaviour. As the simulation progresses, the tail of the distribution rises by several orders of magnitude and extends progressively further out in momentum space, reflecting the ongoing acceleration of electrons into the runaway regime by the induced electric field during the current quench. The extremely narrow structures at low momentum might explain some of the sensitivities observed in the convergence tests and might require adaptations of the momentum space grid structures for future applications.

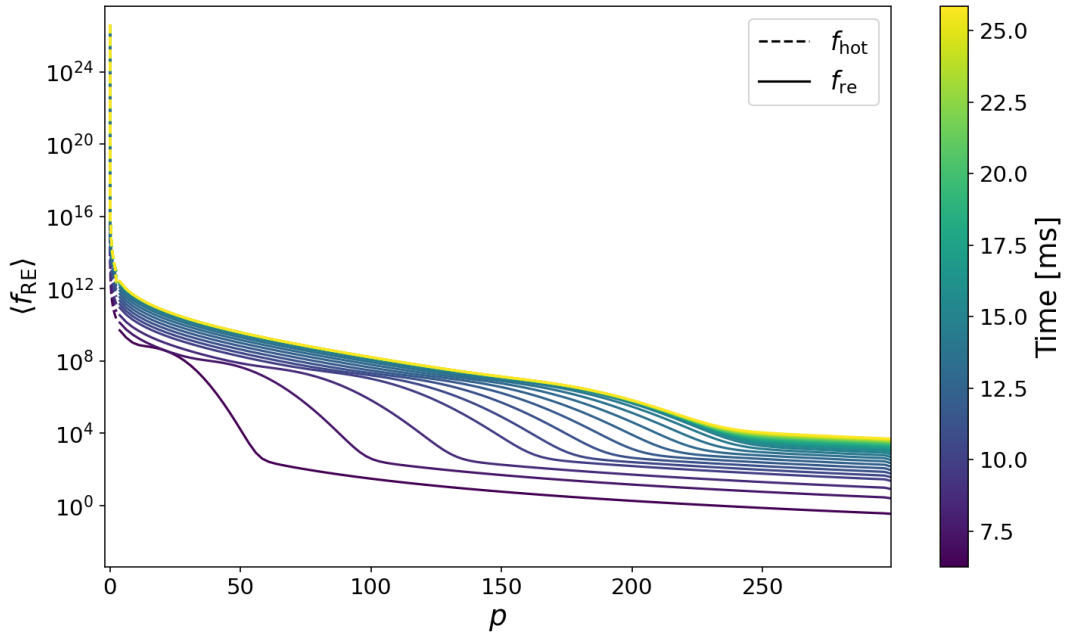


Figure 4.9: Time evolution of the pitch-averaged electron distribution function in DREAM at $r = 0$.

Chapter 5

Conclusion and future work

This thesis has presented a proof of concept for a new modeling framework for runaway electron (RE) generation and dynamics, obtained by a two-way coupling of the JOREK and DREAM codes. Prior to this work, no such coupling existed: JOREK's nonlinear MHD description and DREAM's kinetic, bounce-averaged treatment of the electron distribution function had each been developed and validated independently, but combining the two had not previously been attempted. The coupled model developed here fills exactly this gap: it retains JOREK's accurate free-boundary evolution of the plasma equilibrium, including the vertical motion driving the current quench, while letting DREAM evolve the runaway electron phase-space distribution kinetically, at a computational cost far below that of JOREK's own three-dimensional kinetic module. This opens the perspective for future simulations crossing the enormous time scale of the CQ in large tokamaks like ITER while tracing the plasma vertical motion and the RE kinetic distribution consistently throughout.

To achieve this, several new numerical algorithms were developed and implemented as part of this thesis: an algorithm to track the evolution of magnetic flux surfaces between coupling steps using guiding-centre marker particles, a conservative remapping scheme to transfer the electron distribution functions between successive DREAM equilibria without introducing spurious particle sources or losses, and a scheme to project the DREAM-computed runaway current density back onto JOREK's finite-element grid and incorporate it self-consistently in the induction equation. Each of these algorithms was verified independently, both analytically and numerically, before the fully coupled model was benchmarked against a reference JOREK disruption simulation of the EU-DEMO tokamak using the JOREK fluid RE model. The benchmark comparison showed that the coupled model reproduces the overall current and vertical-displacement dynamics of the reference simulation already with reasonable accuracy, while additionally providing the phase-space resolved electron distribution function throughout the current quench.

At the same time, the convergence studies carried out in this thesis also revealed some open numerical questions that could not be fully resolved within the available time scale and will require a deeper investigation of numerical methods used in JOREK, DREAM, and the coupling itself. In particular, the convergence behaviour with respect to the coupling time step did not show the expected monotonic trend at the finest resolution tested, suggesting that some aspect of the temporal coupling needs further investigation. Similarly, while the radial and momentum-space convergence studies demonstrated satisfactory behaviour for the plasma and runaway currents already at moderate resolution, the magnetic axis position proved more sensitive to resolution and had not yet fully converged at the highest resolutions tested here, primarily due to the steep computational

cost of further increasing the radial resolution in DREAM. Addressing this scaling, for instance through further optimization or parallelization of DREAM's matrix construction and solver routines, as it is presently being in activities independent from this thesis, will allow these convergence studies to be extended accordingly. Furthermore, the end goal of the JOREK-DREAM coupling is to simulate the whole current quench and runaway plateau phases. The plateau phase is numerically challenging to capture due to the presence of near-threshold electrical field values, which require specially adapted numerical methods, highly accurate treatment of collisions, and further adaptations beyond the scope of this thesis.

Looking beyond this thesis project, the coupled model is intended to serve as one important element in a broad hierarchy of RE simulation tools spanning the full disruption event, as outlined in Section 3.1. Realizing this chain in practice will require further development on several fronts. On the physical side, the coupling needs to be extended to include additional kinetic effects that are not yet incorporated in the present implementation, such as drift orbits and the RE influence onto the major-radial force balance. On the numerical side, the independently ongoing efforts to parallelize DREAM's solver routines are expected to substantially reduce the cost of fully kinetic simulations, which will enable running coupled simulations with higher precision and resolution. With these improvements, the JOREK-DREAM coupled model is expected to see broad application in future studies of runaway electron generation and transport over long disruption time scales in next-generation, high-current tokamaks such as ITER and EU-DEMO, where accurately predicting the runaway current and beam characteristics is of direct relevance for machine protection. Finally, validating the coupled model against experimental disruption data, rather than only verifying it against other simulation tools as done in this thesis, is planned as a next step.

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